

Malware Analysis Report For elf-Linux-ia64-bash

Binary Type

ELF (Executable and Linkable Format). (Approved)

Analyzing File

- **File Name:** elf-Linux-ia64-bash
- **File Extension:**
- **File Type:** ELF (Executable and Linkable Format).
- **MD5 Hash:** f1aa2c140a3f56da03a53c6563d1af99
- **SHA256 Hash:** a99374c4a111d7f451bd93ed8aa68df70acd3fbd681c984371ddcd95481775bc

File Identification

File detected as a valid Linux executable.

ELF File Analysis

ELF File Analysis: True

ELF File Imports

- libreadline.so.6
- libncurses.so.5
- libdl.so.2
- libc.so.6.1

ELF File Exports

No Exported functions were found.

String Analysis

ASCII Strings

- /lib/ld-linux-ia64.so.2
- P +"
- 3D1S""
- WWqL_L;G
- "hj+ld
- 47*kP#
- ?;*j2#
- ,2=v1t
- e")O}F
- libreadline.so.6
- **gmon_start**
- JvRegisterClasses
- using_history
- rlmakebare_keymap
- rhreadlinestate
- rlcompletionmarksymlinkdirs
- rlvistart_inserting
- history_length
- history_get
- rlpreterm_function
- rldetetext
- emacsstandardkeymap
- rladdundo
- rlinvokingkeyseqs
- rlbindingkeyseqinmap
- history_tokenize
- rlfilenamequoting_desired
- shunsetnodelay_mode
- rltabinsert
- tildeadditional/suffixes
- rlexplicitarg

- `rlparseand_bind`
- `rl_crlf`
- `remove_history`
- `rlvibWord`
- `history_expand`
- `rlnamedfunction`
- `historysearchdelimiter_chars`
- `rlcompleterwordbreakcharacters`
- `rlterminalname`
- `rl_end`
- `emacsmetakeymap`
- `historygettime`
- `rlfunctionof_keyseq`
- `shgetenv_value`
- `rltildeexpand`
- `rlattemptedcompletion_over`
- `rl_dispatching`
- `rlreadinit_file`
- `vimovementkeymap`
- `rlfilenamedequoting_function`
- `where_history`
- `rlcompletiontype`
- `rldirectorycompletion_hook`
- `rlfunmapnames`
- `tildeexpansionpreexpansion_hook`
- `rleditingmode`
- `rl/astfunc`
- `shsinglequote`
- `rlfilenamequoting_function`
- `historyquotesinhibit_expansion`
- `shsetlinesandcolumns`
- `history_base`
- `rlforwardword`
- `rlcharisquotedp`
- `rlcompletionquote_character`
- `rl_done`
- `rl_initialize`
- `rlbindkeyinmap`
- `rlunbindfunctioninmap`
- `rlcompletionentry_function`
- `rladddfun`
- `rlcompletionssuppress_quote`
- `rl_instream`
- `rlnumcharstoread`
- `rlgetprevious_history`
- `rlsetkeymap`
- `historysubstchar`
- `rlcompletionappend_character`
- `rlpushmacro_input`
- `rlignoresomecompletionsfunction`
- `rlvariablebind`
- `write_history`
- `rlunbindkeyinmap`
- `historyinhibitexpansion_function`
- `rlgenericbind`
- `unstifle_history`
- `rlattemptedcompletion_function`
- `rlforcedupdate_display`
- `rlcompletioninvoking_key`
- `rlbindkeyifunboundinmap`
- `shgethomedir`
- `rlsortcompletion_matches`
- `rlfilenamerewrite_hook`
- `rlvariabledumper`
- `rl_outstream`
- `rl/istfunmap_names`
- `historyisstified`
- `rlbindkeyseq`
- `rlcompletionmode`
- `rlexecutingkeymap`
- `rlcompleterquote_characters`

- *rlfilenamecompletion_desired*
- *emacsct/xkeymap*
- *freehistoryentry*
- *rl_insert*
- *rlreadkey*
- *historywritetimestamps*
- *rlgetkeymapbyname*
- *rlfilenamecompletion_function*
- *tildeadditionalprefixes*
- *historycommentchar*
- *rlcompleteinternal*
- *rlspecialprefixes*
- *rllinebuffer*
- *rlonnew_line*
- *rlusernamecompletion_function*
- *rl_newline*
- *rl_complete*
- *rlviediting_mode*
- *rlfunctiondumper*
- *rlgetkeymap*
- *rlinserttext*
- *viinsertionkeymap*
- *rlvariablevalue*
- *rlreadlinename*
- *rlcompletionmatches*
- *rlstartuphook*
- *historytruncatefile*
- *rl_redisplay*
- *read_history*
- *rlbasicwordbreakcharacters*
- *rl_point*
- *rlmenucompletionentryfunction*
- *rlmacroddumper*
- *rl_yanklast_arg*
- *historysetpos*
- *replacehistoryentry*
- *rlresetterminal*
- *rlviend_word*
- *rl_mark*
- *rlsetscreen_size*
- *rlkilltext*
- *rlcompletionfound_quote*
- *readhistoryrange*
- *rldeprepterm_function*
- *historymaxentries*
- *historyexpansionchar*
- *rlfilenamequote_characters*
- *rldirectoryrewrite_hook*
- *rlmenucomplete*
- *rl_ding*
- *rlgettermcap*
- *rlcompletionssuppress_append*
- *libncurses.so.5*
- *libdl.so.2*
- *dlopen*
- *dclose*
- *derror*
- *libc.so.6.1*
- *setuid*
- *strcasestr*
- *socket*
- *fflush*
- *strcpy*
- *__printfchk*
- *setgrent*
- *readdir*
- *pathconf*
- */Oputc*
- *setlocale*
- *mbrtowc*
- *towlower*

- gai_strerror
- strsignal
- strncmp
- dcgettext
- setservent
- strchr
- *_longjmpchk*
- regexec
- dcgettext
- wctomb
- getpwuid
- *_fdeltchk*
- closedir
- strncpy
- sigprocmask
- regfree
- getgrent
- __lxstat
- unlink
- putchar
- select
- strtold
- towupper
- getpid
- strftime
- iswalnum
- iswlower
- *_ctypegetmbcur_max*
- endpwent
- strtol
- isatty
- strchmul
- getppid
- wcschr
- iswctype
- settimer
- sigemptyset
- getaddrinfo
- memset
- strstr
- *_ermolocation*
- tcsetattr
- confstr
- endgrent
- mbsnrtowcs
- mbtowc
- ttyname
- clearerr
- unsetenv
- *_fprintfchk*
- sigaddset
- iswupper
- putenv
- wcswidth
- __sigsetjmp
- stdout
- getrusage
- strlen
- memcpy
- fclose
- tcsetpgrp
- *_memmovechk*
- wcscmp
- *_strtolinternal*
- *_vsnprintfchk*
- strtoul
- killpg
- strcat
- strcasecmp
- setpgid
- getpeername
- mbsinit
- *_strncpychk*

- `getgid`
- `opendir`
- `_ctypeb_loc`
- `regcomp`
- `stderr`
- `sigdelset`
- `group_member`
- `_snprintfchk`
- `getgroups`
- `getuid`
- `readlink`
- `getegid`
- `setrlimit`
- `__fpurge`
- `strncasecmp`
- `__fxstat`
- `fileno`
- `gethostname`
- `getcwd`
- `fwrite`
- `gettimeofday`
- `sigaction`
- `geteuid`
- `_memcpychk`
- `waitpid`
- `localtime`
- `strchr`
- `_vfprintfchk`
- `fdopen`
- `tcgetattr`
- `_ctypetoupper_loc`
- `_strncpychk`
- `_ctypetolower_loc`
- `freeaddrinfo`
- `setvbuf`
- `wcscoll`
- `_sprintfchk`
- `wcsdup`
- `__xstat`
- `getrlimit`
- `getdtablesize`
- `memmove`
- `endservent`
- `bindtextdomain`
- `wcsrtombs`
- `setgid`
- `faccessat`
- `_asprintfchk`
- `tcgetpgrp`
- `_libcstart_main`
- `ferror`
- `stpcpy`
- `wcslen`
- `strcoll`
- `wcsncmp`
- `_strtolinternal`
- `sysconf`
- `getservent`
- `wctomb`
- `__environ`
- `_edata`
- `_bssstart`
- `sigwinch_sighandler`
- `sh_modcase`
- `parse_shellopts`
- `parse_string`
- `protected_mode`
- `_libccsu_fini`
- `searchforcommand`
- `xdupmbstowcs`
- `shellglobfilename`
- `clean_itemlist`
- `declare_doc`

- `countalljobs`
- `commandwordcompletion_function`
- `arraydequoteescapes`
- `find_variable`
- `running_trap`
- `break_doc`
- `jobs_doc`
- `dirspell`
- `coproc_fdrestore`
- `fmtulong`
- `shopt_setopt`
- `arraydisposeelement`
- `set_shellopts`
- `pop_args`
- `lastpipe_opt`
- `noexitonfailedexec`
- `shouldignoreglob_matches`
- `duplicatebufferedstream`
- `bashline_reset`
- `getifs`
- `executing_builtin`
- `all_digits`
- `tempvar_list`
- `completionstoringlist`
- `getcurrentprompt_level`
- `pat_subst`
- `addorsupercedeexportedvar`
- `shownameattributes`
- `toplevelmask`
- `bashinitializehistory`
- `theprintedcommand`
- `addunwindprotect`
- `var_lookup`
- `fd_ispipe`
- `nosymboliclinks`
- `strvecfromword_list`
- `expandstringunsplittostring`
- `resetmailtimer`
- `theprintedcommandexcepttrap`
- `setdefaultlocale`
- `makewhilecommand`
- `assoc_patsub`
- `executinglinenumber`
- `checkdevtty`
- `resetshelloptions`
- `buffered_ungetchar`
- `ynerrs`
- `cleanupthepipeline`
- `wmatchlen`
- `disposefunctiondef_contents`
- `restricted_shell`
- `terminate_immediately`
- `hash_doc`
- `zwrite`
- `array_shift`
- `signalistrapped`
- `bashildeexpand`
- `ulimit_builtin`
- `logout_doc`
- `getdirstackfrom_string`
- `zmapfd`
- `select_doc`
- `dotordotdot`
- `wdcache`
- `gatherheredocuments`
- `print_rlimtype`
- `dstack`
- `trim_pathname`
- `rundebugtrap`
- `closefdbitmap`
- `break_builtin`
- `sv_histtimefmt`

- `closebufferedstream`
- `bashgetcwderrstr`
- `breaking`
- `historylinesin_file`
- `glob_star`
- `convertvarto_array`
- `bash_copyright`
- `subshell_envp`
- `remembermaildates`
- `fg_doc`
- `builtin_error`
- `builtin_address`
- `remove_backslashes`
- `freemailfiles`
- `parser_error`
- `makevariablevalue`
- `command_errstr`
- `coproc_fdchk`
- `mailstat`
- `expandwordsno_vars`
- `u32cconv`
- `it_arrayvars`
- `reader_loop`
- `localexportedvariables`
- `sh_syntabsiz`
- `coproc_flush`
- `make_command`
- `local_builtin`
- `hostname/istinitialized`
- `findpathfile`
- `assoc_insert`
- `file_exists`
- `zcatfd`
- `caller_doc`
- `expandwordsshellexp`
- `whichsetflags`
- `ifs_firstc`
- `strcreplace`
- `caller_builtin`
- `bindkeyseqtounixcommand`
- `executevariablecommand`
- `builtin_warning`
- `exitimmediatelyon_error`
- `findstringin_alist`
- `push_context`
- `sigint_sighandler`
- `primary_prompt`
- `mbschr`
- `redirection_expand`
- `hash_builtin`
- `shell_initialized`
- `tt_setnocanon`
- `fgpercentdoc`
- `bash_logout`
- `closebufferedfd`
- `ttstore`
- `ifs_cmap`
- `disown_builtin`
- `findomakearrayvariable`
- `arrayremovequoted_nulls`
- `sh_stat`
- `continue_builtin`
- `globignorecase`
- `strlist_resize`
- `coproc_setvars`
- `bgpids`
- `disableprivmode`
- `falarm`
- `array_rshift`
- `shellversionstring`
- `xbcopy`
- `strlist_flush`

- `posixreadlineinitialize`
- `printarithcommand`
- `test_command`
- `suspend_builtin`
- `fnx_tofs`
- `isbasictable`
- `coproc_fdsave`
- `expandpromptstring`
- `groupingbracesdoc`
- `xtrace_reset`
- `setsiginthandler`
- `noemptycommand_completion`
- `base_pathname`
- `allocworddesc`
- `defaultbufferedinput`
- `addtimeval`
- `makecoproccommand`
- `arrayexpandindex`
- `aliasexpandword`
- `secondary_prompt`
- `array_modcase`
- `liststoppedjobs`
- `string/listdollar_star`
- `popvarcontext`
- `enablehostnamecompletion`
- `push_scope`
- `maybeexport_env`
- `arrayshiftelement`
- `copyfunctiondef_contents`
- `push_stream`
- `getminuso_opts`
- `echo_builtin`
- `sh_optopt`
- `buffered_getchar`
- `return_EOF`
- `currentfdsto_close`
- `bg_doc`
- `zgetline`
- `exit_builtin`
- `let_doc`
- `skiptodelim`
- `signalisspecial`
- `notifyandcleanup`
- `make_word`
- `builtin_doc`
- `dotfoundin_search`
- `termsig_sighandler`
- `read_command`
- `ttfd_cbreak`
- `build_version`
- `exit_doc`
- `savedcommandline_count`
- `currentcommandline_count`
- `copy_command`
- `expandwordunsplit`
- `dowordassignment`
- `findshellbuiltin`
- `makebareword`
- `setupglobignore`
- `readsecondaryline`
- `shelleoftoken`
- `dircompleteexpandrelpath`
- `currentreadline`
- `initialize_readline`
- `copyfunctiondef`
- `popd_doc`
- `yyerror`
- `getlocalevar`
- `unsetbashinput`
- `returncatchvalue`
- `assocdequoteescapes`
- `yparse`

- waitforjob
- svstrictposix
- array_patsub
- makeifcommand
- it_running
- times_doc
- legalaliasname
- sh_regmatch
- currentcommandfirstlinesaved
- disposefunctiondef
- allvisiblefunctions
- subshell_level
- parse_bashopts
- bashbadsuberrmsg
- complete_builtin
- list_string
- ungetcwithrestart
- setsignalhandler
- alreadymakingchildren
- lastmadepid
- zreadintr
- word_split
- findspecialbuiltin
- sh_optind
- waitforbackground_pids
- wordexp_only
- sh_eaccess
- shellcompatibilitylevel
- quotestringfor_globbing
- sv_mail
- withinputfrom_string
- lcurrent
- makelocalarray_variable
- bindintvariable
- setloginshell
- pushedstringlist
- ifs_var
- findusercommand
- sh_realpath
- fmtumax
- arraytoassign
- pcomp_curcmd
- nohup_job
- setworkingdirectory
- lastcommandexit_signal
- shellisrestricted
- makefunctiondef
- times_builtin
- shell_tty
- shbackslashquote
- strlist_print
- savebashinput
- set_signal
- maybesetsigchild_trap
- initializeshelloptions
- hash_dispose
- getnamefor_error
- findindexin_alist
- readonly_builtin
- assignarrayelement
- describe_pid
- unwindprotectmem
- getworkingdirectory
- setifnot
- failglobexpansion
- localetrans
- assignarrayfrom_string
- sv_optind
- return_doc
- sv_opterr
- terminatestoppedjobs
- dircomplete_expand

- `setdefaultlocale_vars`
- `sv_funcnest`
- `svcompwordbreaks`
- `commandstringindex`
- `shbackslashquotefordouble_quotes`
- `pcomp_curcs`
- `ttnocanon`
- `sh_nojobs`
- `until_doc`
- `set_pwd`
- `findfunctiondef`
- `phash_remove`
- `gencompspeccompletions`
- `source_builtin`
- `killalllocal_variables`
- `namedfunctionstring`
- `tt_settonechar`
- `setpipestatusarray`
- `getnumericarg`
- `internal_warning`
- `hash_copy`
- `expandstringto_string`
- `tt_setcbreak`
- `listminuso_opts`
- `findtempenvvariable`
- `subshelltoplevel`
- `sh_erange`
- `pop_context`
- `withinputfrom_stream`
- `expand_string`
- `assoctoassign`
- `return_catch`
- `it_jobs`
- `array_quote`
- `netopen`
- `stringlistdollar_at`
- `it_keywords`
- `array_dispose`
- `glob_vector`
- `change_flag`
- `trap_doc`
- `unset_builtin`
- `initializeshell/builtins`
- `dequote_list`
- `bind_function`
- `unbind_func`
- `expandwordleave_quoted`
- `setsignalignored`
- `bashline_reinitialize`
- `print_command`
- `parsesymbolicmode`
- `xtrace_fp`
- `initialize_signals`
- `stringrestof_args`
- `signal_names`
- `xrealloc`
- `ps2_prompt`
- `EOF_Reached`
- `sv_terminal`
- `makenewarray_variable`
- `wordtokenalist`
- `check_identifier`
- `makeuntilcommand`
- `command_connect`
- `bashaddhistory`
- `ttfd_noecho`
- `theprintedcommand_size`
- `unquotedglobpattern_p`
- `resetshoptoptions`
- `pushvarcontext`
- `makecondcommand`
- `sccs_version`

- sh_syntaxtab
- prog_completes
- programming_error
- savepipestatusarray
- substring
- jobexitsignal
- newvarcontext
- hash_string
- unlinkfifolist
- historylinesthis_session
- sh_physpath
- jobexitstatus
- progcomp_remove
- sh_mktmpfd
- sourcesearchescwd
- setjobcontrol
- getwordfrom_string
- printvarvalue
- map_over
- setsigchldhandler
- reinitspecialvariables
- fcexecutefile
- unbindfunctiondef
- logout_builtin
- printshifterror
- fifos_pending
- copyfifolist
- export_builtin
- openbufferedstream
- huponexit
- coproc_list
- resetsignalhandlers
- it_users
- push_args
- initializeshellvariables
- setimpossiblesigchld_trap
- skipsubscript
- tempenvvassigerror
- echo_doc
- phash_flush
- arrayfromword_list
- array_copy
- wait_doc
- extractarithmeticsubst
- xmalloc
- yyinputname
- killcurrentpipeline
- eval_builtin
- dircomplete_spelling
- convertvarto_assoc
- hist_verify
- internal_wstrmatch
- maybeexecutefile
- strvec_strcmp
- restartjobcontrol
- filter_stringlist
- compspec_copy
- compgen_doc
- makunbound
- phash_search
- thisshellfunction
- historyexpansioninhibited
- interrupt_immediately
- array_remove
- minusoption_value
- syncbufferedstream
- pwd_builtin
- absolute_program
- progcomp_dispose
- ttfd_onechar
- runtrapcleanup
- getgrouparray

- `decodepromptstring`
- `mail_warning`
- `arithfdoc`
- `xtrace_init`
- `staticshell/builtins`
- `setcurrentprompt_level`
- `makebaresimple_command`
- `shdoublequote`
- `subshell_argv`
- `zreset`
- `currentreadlineline_index`
- `eof_encountered`
- `umatchlen`
- `assocquoteescapes`
- `dot_doc`
- `readbutdont_execute`
- `phash_create`
- `source_file`
- `setreturntrap`
- `hash_size`
- `loop_level`
- `cdable_vars`
- `privileged_mode`
- `ifs_value`
- `builtin_usage`
- `executable_file`
- `expandassignmentstringtostring`
- `bindvarto_int`
- `shell_name`
- `remove_alias`
- `setdirectoryhook`
- `allarrayvariables`
- `giveternalto`
- `justonecommand`
- `trapsavedexit_value`
- `arraycreateelement`
- `command_error`
- `allownullglob_expansion`
- `true_doc`
- `getcoprocbypid`
- `assignarrayvarfromword_list`
- `make_newassoc_variable`
- `debuggingloginshell`
- `getnewwindow_size`
- `sh_mktmpfp`
- `pipefail_opt`
- `withinputfrom_stdin`
- `builtin_builtin`
- `wantpendingcommand`
- `ps1_prompt`
- `interrupt_state`
- `return_builtin`
- `displaysignalist`
- `xtrace_fd`
- `copywordlist`
- `stringlistinternal`
- `compspec_create`
- `ttsave`
- `redirection_error`
- `restorepgppipe`
- `setvarattribute`
- `complete_doc`
- `read_doc`
- `dispose_variable`
- `variable_context`
- `runsigchldtrap`
- `strlist_create`
- `coproc_wclose`
- `_datastart`
- `strlisttoword_list`
- `makegroupcommand`
- `setminuso_option`

- `executeshell/function`
- `waitsignal/received`
- `setbashinput_fd`
- `hash_walk`
- `clearhostnamelist`
- `wait_builtin`
- `setbufferedstream`
- `strvectoword_list`
- `rununwindframe`
- `dispose_command`
- `internal_strmatch`
- `extractcolonunit`
- `promptvars`
- `compspec_dispose`
- `strlist_walk`
- `kill_doc`
- `sh_notbuiltin`
- `placekeywordsin_env`
- `expandstringassignment`
- `assoc_quote`
- `do_redirections`
- `sh_invalidoptname`
- `extractcommandsubst`
- `disallowfilenameeglobbing`
- `initializejobsignals`
- `print_timeval`
- `colon_doc`
- `timevaltocpu`
- `exec_argv0`
- `it_hostnames`
- `coproc_init`
- `setpipestatusfrom_exit`
- `matchpatternwchar`
- `kill_pid`
- `test_builtin`
- `histlastline_added`
- `freepushedstring_input`
- `seterrortrap`
- `adjustshelllevel`
- `updateexportenv_inplace`
- `unbind_variable`
- `sv_ifs`
- `popdollarvars`
- `_libcia64registerbackingstorebase`
- `xtrace_set`
- `coproc_closeall`
- `functiontracemode`
- `sh_opterr`
- `it_commands`
- `xparse_dolparen`
- `restoreoriginalsignals`
- `sh_coproc`
- `brace_expansion`
- `expand_words`
- `subappendstring`
- `ttgetattr`
- `ansicstr`
- `quoteglobberchars`
- `inityyio`
- `do_assignment`
- `resetmailfiles`
- `zreadc`
- `shopt_doc`
- `dumpppostings`
- `shift_doc`
- `parseandexecute_cleanup`
- `getmaxgroups`
- `resetterminatingsignals`
- `wordlistremovequotednulls`
- `posix_initialize`
- `bind_doc`
- `enable_doc`

- `comptopt_builtin`
- `setoriginalsignal`
- `bashhistorydisable`
- `subshell_environment`
- `printf_doc`
- `disposefdbitmap`
- `withoutjobcontrol`
- `getarrayvalue`
- `striptrailingifs_whitespace`
- `maybecalltrap_handler`
- `shellstarttime`
- `promptstringpointer`
- `typeset_doc`
- `exec_builtin`
- `forced_interactive`
- `malloctraceat_exit`
- `phash_insert`
- `sourceusespath`
- `indirectionlevelstring`
- `listalljobs`
- `enable_builtin`
- `/Ostdin_used`
- `strvec_len`
- `stream_list`
- `shundouble_quote`
- `asynchronous_notification`
- `deleteallcontexts`
- `setlocalevar`
- `patch_level`
- `signal_name`
- `makefuncnamevisible`
- `absolute_pathname`
- `append_process`
- `getalloriginal_signals`
- `lastasynchronouspid`
- `assoctoword_list`
- `yychar`
- `getopts_doc`
- `it_aliases`
- `arrayvariablename`
- `findvariableinternal`
- `local_doc`
- `printarrayassignment`
- `coproc_pidchk`
- `setdollarvars_unchanged`
- `sh_validfd`
- `stringfortimtype`
- `command_substitute`
- `maybe-makerestricted`
- `function_doc`
- `err_unboundvar`
- `it_directories`
- `sh_badjob`
- `resetcompleterwordbreakchars`
- `arrayquoteescapes`
- `unbind_args`
- `coproc_close`
- `listrestof_args`
- `bashdirectorycompletion_matches`
- `expandarithstring`
- `currentreadlineprompt`
- `literal_history`
- `strsub`
- `interactive_shell`
- `while_doc`
- `eofencounteredlimit`
- `cleansimplecommand`
- `arith_doc`
- `sh_closepipe`
- `arrayassignlist`
- `sh_invalidid`
- `checkhashedfilenames`

- sh_exit
- strlist_append
- sys_error
- dispose_redirects
- programmable_completions
- makecommandstring
- unfreezejoblist
- fatal_error
- ifs_firstchar
- suspend_doc
- disposeworddesc
- saveparserstate
- make_absolute
- currentpromptstring
- array_walk
- makearithfor_command
- getmaxchild
- bashildefind_word
- sh_badpid
- strvec_copy
- sh_getopt
- arraytoword_list
- assoc_flush
- case_doc
- fsleep
- traptosighandler
- mapfile_builtin
- getvariablevalue
- dumptranslatablestrings
- uinttostr
- ansiexpand
- signalinprogress
- dispose_words
- getshoptoptions
- compgen_builtin
- assoc_remove
- timevaltosecs
- dollardollarpid
- getdollarvar_value
- nolinediting
- current_token
- binary_test
- assockeystowordlist
- setdefaultlang
- _libccsu_init
- needheredoc
- internal_error
- jobs_builtin
- file_isdir
- runpendingtraps
- ifsfirstclen
- compopt_doc
- full_pathname
- setupignorepatterns
- listonejob
- type_builtin
- matchignorecase
- pop_scope
- maybeappendhistory
- bashreedit
- assoc_dispose
- showalvar_attributes
- makedefaultmailpath
- deletealljobs
- currentcommandsubst_pid
- haveunwindprotects
- num_fifos
- it_disabled
- sh_invalidopt
- getstringvalue
- eval_doc
- bashhistoryreinit

- disown_doc
- strvec_create
- exec_doc
- get_exitstat
- saveinputline_state
- initjobstats
- markmodifiedvars
- history_reediting
- validarrayreference
- current_user
- false_builtin
- cdspelling
- stringlistpos_params
- allexportedvariables
- putcommandnameintoenv
- sh_invalidsig
- setuphistoryignore
- array_insert
- ttnoecho
- bind_builtin
- currentcommandnumber
- dirs_builtin
- false_doc
- enablehistorylist
- arraykeystowordlist
- it_builtins
- commandexecutionstring
- setorshow_attributes
- bindvariablevalue
- assocremovequoted_nulls
- setdebugtrap
- current_builtin
- condexpandword
- runreturntrap
- posixly_correct
- global_variables
- it_functions
- findglobalvariable
- report_error
- sh_openpipe
- xstrmatch
- printfundist
- checkbashinput
- array_dequote
- removequotedescapes
- makeselectcommand
- sigwinch_received
- ttattr
- set_ppid
- timetocheck_mail
- xpg_echo
- xtraceprintword_list
- file_iswdir
- errortracemode
- it_services
- trap_list
- listrunningjobs
- let_builtin
- type_doc
- export_doc
- sh_neednumarg
- execredirectundo_list
- progcomp_search
- svhistorycontrol
- global_command
- wlcach
- assoctostring
- jobsmflag
- pop_stream
- it_signals
- wcsmatch
- cmd_init

- coproc_checkfd
- strvec_sort
- mk_msgstr
- ttonechar
- find_alias
- alias_expand
- zreadretry
- bindarrayvariable
- ulimit_doc
- resetinternalgetopt
- rememberonhistory
- globerrorreturn
- progcomp_flush
- sh_makepath
- progcomp_insert
- disposesaveddollar_vars
- initializejobcontrol
- autocd
- restore_pipeline
- it_variables
- colon_builtin
- err_readonly
- xtraceprintassignment
- fc_builtin
- interactive_comments
- xtrace_fdchk
- runinterrupttrap
- makesimplecommand
- file_status
- list_reverse
- sv_xtracefd
- stringquoteremoval
- bash_license
- nohupalljobs
- performhostnamecompletion
- clearunwindprotect_list
- jumptotop_level
- xtraceprintforcommandhead
- terminatecurrentpipeline
- settystate
- comsubignorereturn
- xtraceprintarith_cmd
- u32tochar
- xtraceprintselectcommandhead
- unalias_doc
- arrayvariablepart
- noinvisiblevars
- setupasynsignals
- showsshellversion
- funcnest_max
- streamonstack
- tilde_initialize
- it_enabled
- bg_builtin
- history_number
- preprocessline
- readonly_doc
- list_optarg
- initialize_traps
- copy_redirects
- sv_histchars
- umask_builtin
- makepatternlist
- shopt_builtin
- no_options
- tteightbit
- list_optopt
- deleteallaliases
- if_doc
- testbracketdoc
- restoreparserstate
- returncatchflag

- `closepgppipe`
- `commandorientedhistory`
- `getcurrentuser_info`
- `reapdeadjobs`
- `savetokenstate`
- `pcompsetreadline_variables`
- `pwd_doc`
- `sh_mktmpname`
- `beginunwindframe`
- `strlist_remove`
- `release_status`
- `sv_histsize`
- `isnetconn`
- `splitatdelims`
- `array_create`
- `makebuiltinargv`
- `setvarread_only`
- `zreadcn`
- `arrayunshiftelement`
- `initialize_bashopts`
- `executecommandinternal`
- `terminating_signal`
- `time_doc`
- `getcoprocbname`
- `initializebashinput`
- `umask_doc`
- `makesubshellcommand`
- `cd_doc`
- `closenewffios`
- `u32toutf8`
- `strlistprefixsuffix`
- `sv_globignore`
- `findreservedword`
- `findtokenin_alist`
- `lasthistoryline`
- `parseandexecute`
- `printvarlist`
- `readarray_doc`
- `ansic_quote`
- `arraytoargv`
- `readfromstdin`
- `gnuerrorformat`
- `sh_chkwrite`
- `maybesaveshell_history`
- `strvec_flush`
- `ttsetattr`
- `makeheredocument`
- `coproc_dispose`
- `history_doc`
- `xtraceprintcond_term`
- `hashing_enabled`
- `ttfd_nocanon`
- `savepgppipe`
- `dontsavefunction_defs`
- `builtinaddressinternal`
- `coproc_rclose`
- `fc_doc`
- `ignore_signal`
- `shift_builtin`
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- `bashhistoryenable`
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- `dirs_doc`
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- `fdisbash_input`
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- *echoinputat_read*
- *movetohigh_fd*
- *strlist_dispose*
- *makearithcommand*
- *noglobdotfilenames*
- *setifs*
- *stupidlyhackspecial_variables*
- *shellfunctiondefs*
- *showvarattributes*
- *matchpatternchar*
- *thisshellbuiltin*
- *load_history*
- *checkaddhistory*
- *continuing*
- *stdin_redirects*
- *fg_builtin*
- *printassocassignment*
- *substassignvarlist*
- *termsig_handler*
- *force_ignore*
- *exit_shell*
- *sh_badopt*
- *sv_hostfile*
- *arraytostring*
- *indirection_level*
- *hash_bucket*
- *lastcommandsubst_pid*
- *strlist_sort*
- *save_pipeline*
- *lastshellbuiltin*
- *mbstrlen*
- *it_exports*
- *continue_doc*
- *parseandexecute_level*
- *strip_trailing*
- *initialize_itemlist*
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- *is_directory*
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- *mapfile_doc*
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- *copy_word*
- *test_unop*
- *strvec_remove*
- *arrayneedsmaking*
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- *parse_command*
- *terminal_pgrp*
- *checkbinaryfile*
- *rununwindprotects*
- *legal_number*
- *sh_setlinebuf*
- *makewordfrom_token*
- *it_stopped*
- *reset_parser*
- *brace_expand*
- *charisquoted*
- *localeexpand*
- *printf_builtin*
- *forceappendhistory*
- *unsetsigwinchhandler*
- *remember_args*
- *set_bashopts*
- *progcomp_create*
- *printcondcommand*
- *expandcompoundarray_assignment*
- *tt_seteightbit*
- *coproc_doc*
- *sh_werror*
- *initmaildates*
- *runexittrap*

- `getopts_builtin`
- `dispose_word`
- `test_binop`
- `shcontainsshell metas`
- `toplevelcleanup`
- `disposevarcontext`
- `currenthostname`
- `disposeconddnode`
- `thiscommandname`
- `coproc_unsetvars`
- `sh_restricted`
- `makewordflags`
- `endjobcontrol`
- `unbindarrayelement`
- `assoc_modcase`
- `signalishard_ignored`
- `setitemlistdirty`
- `strlistfromword_list`
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- `read_builtin`
- `getdirectorystack`
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- `decode_signal`
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- `assignarrayvarfromstring`
- `yylval`
- `input_avail`
- `doassignmentno_expand`
- `chkexport`
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- `expandstringunsplit`
- `bashbracecompletion`
- `sv_locale`
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- `zsyncfd`
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- `pushd_doc`
- `allvariablesmatching_prefix`
- `sh_optarg`
- `_getenv`
- `it_setopts`
- `mbsmbchar`
- `waitintrbuf`
- `tvlist_ind`
- `err_badarraysub`
- `execute_command`
- `bashdeletelast_history`
- `same_file`
- `test_doc`
- `makeocalvariable`
- `assigncompoundarray_list`
- `checkjobsat_exit`
- `read_octal`
- `deleteallvariables`
- `assigninginenvironment`
- `unclosed_pair`
- `unalias_builtin`
- `progcomp_walk`
- `set_lang`
- `ttcbreak`
- `getjobspec`
- `throwtotop_level`
- `describe_command`
- `assoc_reference`
- `extractarrayassignment_list`
- `sourcelevel`
- `source_doc`
- `make_redirection`
- `sort_variables`
- `sh_mkdoubledquoted`
- `getjobby_pid`

- `diffimeval`
- `strvec_dispose`
- `setdollarvars_changed`
- `hangupalljobs`
- `find_flag`
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- `ignorettyjob_signals`
- `uconvert`
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- `pushdollarvars`
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- `bashclearhistory`
- `pipeline_pgrp`
- `othertokenalist`
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- `printvarfunction`
- `executingcommandbuiltin`
- `bashreadlineinitialized`
- `sh_needarg`
- `histlastline_pushed`
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- `unset_doc`
- `pushd_builtin`
- `sv_ignoreeof`
- `makeforcommand`
- `file_error`
- `getaliasvalue`
- `no_args`
- `de_backslash`
- `original_pgrp`
- `xtraceprintcasecommandhead`
- `initialize_flags`
- `echocommandat_execute`
- `list_length`
- `aliasexpandall`
- `removeunwindprotect`
- `array_flush`
- `numberofargs`
- `expanding_redir`
- `delete_job`
- `trap_builtin`
- `help_builtin`
- `getjobby_name`
- `sh_ttyerror`
- `optflags`
- `command_doc`
- `debugging_mode`
- `bashservicenamecompletion_function`
- `restoreinputline_state`
- `thecurrentmaintainer`
- `runerrortrap`
- `gethostnameelist`
- `extended_glob`
- `sv_path`
- `coproc_alloc`
- `printsimplecommand`
- `restoretokenstate`
- `makecondnode`
- `ansic_shouldquote`
- `makewordlist`
- `spname`
- `getttystate`
- `conditional_doc`
- `legal_identifier`
- `variablehelpdoc`
- `stopmakingchildren`
- `wait_for`
- `newfdbitmap`
- `thecurrentworking_directory`
- `stop_pipeline`

- start_pipeline
- assoc_subrange
- history_expansion
- sv_histignore
- extractprocesssubst
- fdtobuffered_stream
- maybeaddhistory
- it_shopts
- lastcommandexit_value
- array_slice
- allshellvariables
- sh_canonpath
- getopts_reset
- executableordirectory
- history_builtin
- shell_pgrp
- parsestringtowordlist
- getgrouplist
- find_function
- strlist_merge
- bashdequotetext
- unary_test
- historydelimitingchars
- array_subrange
- allvisiblevariables
- coproc_fdclose
- progcompletionenabled
- pcompsetcompspec_options
- progcomp_size
- make_child
- startup_state
- bashgroupnamecompletion_function
- strlist_copy
- sh_notfound
- zreadcintr
- allshellfunctions
- resetshellflags
- makeocalassoc_variable
- coproc_reap
- defaultttyjob_signals
- shell_execve
- start_job
- signal/signored
- xmbsrtowcs
- numshell/builtins
- cd_builtin
- popd_builtin
- tt_setnoecho
- globalerrorlist
- withinputfrombufferedstream
- stdin_redir
- ttfd_eightbit
- waitforsingle_pid
- freebufferedstream
- sh_invalidnum
- initializeterminatingsignals
- assigninenv
- dist_version
- bindassocvariable
- mapoverfuncs
- connectasynclist
- sh_readonly
- mergetemporaryenv
- fnx_fromfs
- bashdelelethist
- it_bindings
- shgetoptrestore_state
- unlink_fifo
- restoredefaultsignal
- bashdefaultcompletion
- restorepipestatusarray
- expand_aliases

- add_alias
- kill_builtin
- linenumbase
- disposeexecredirects
- dequote_string
- shopt_listopt
- discardunwindframe
- GLIBC_2.1
- GLIBC_2.0
- GLIBC_2.11
- GLIBC_2.8
- GLIBC_2.15
- GLIBC_2.4
- GLIBC_2.3.4
- GLIBC_2.2
- GLIBC_2.3
- `@&(!
- 0!L. p
- p@E?#p
- 5t0
- Pffff
- @ffff
- Qffff
- affff
- tH 2*A
- F. ``9
-]. `x9
- < #@\$"E
- E\$@@@\$"E
- ! B80(
- !@qP\$@
- 1D, `
- @80+ G
- u?#`p@
- 9 A< (
- ZD UJ9
- ! B80(
- p B, p:9
- Pffff
- y<"@
- 0AL. @
- 0!L. @
- pXK?#`
- 2(@C mP
- pXK?#`
- !@x8 +
- =0#@\$:
- =0#@\$:
- x80+@
- @ffff
- px!`?#p
- yffff
- Wffff
- yffff
- Wffff
- GNU bash, version %s-(%s)
- ia64-unknown-linux-gnu
- Shell options:
- -irsD or -c command or -O shopt_option
- (invocation only)
- -%s or -o option
- Type `%s -c "help set"' for more information about shell options.
- Type `%s -c help' for more information about shell builtin commands.
- Use the `bashbug' command to report bugs.
- Usage:
- %s [GNU long option] [option] ...
- %s [GNU long option] [option] script-file ...
- GNU long options:
- /usr/share/bashdb/bashdb-main.inc
- runonecommand
- BASH_ENV
- I have no name!
- ??host??

- POSIXLY_CORRECT
- POSIX_PEDANTIC
- \s-\W\$
- ~/.bashrc
- %s: option requires an argument
- %s: invalid option
- %c%c: invalid option
- login_shell
- SSH_CLIENT
- SSH2_CLIENT
- /etc/bash/bashrc
- /etc/profile
- ~/.profile
- ~/.bash_profile
- ~/.bash_login
- FUNCNAME
- BASH_SOURCE
- BASH_LINENO
- %s: is a directory
- %s: cannot execute binary file
- debugger
- dump-po-strings
- dump-strings
- init-file
- noediting
- noprofile
- protected
- rcfile
- restricted
- verbose
- version
- timed out waiting for input: auto-logout
- PROMPT_COMMAND
- reader_loop
- readline stdin
- %a %b %d
- %H:%M:%S
- %l:%M:%S
- %l:%M %p
- unexpected EOF while looking for matching `%'`
- #%^,~,:=?+/
syntax error near unexpected token `%'`
syntax error near `%'`
syntax error: unexpected end of file
syntax error
unexpected EOF while looking for matching `)`
unexpected EOF while looking for `]]`
syntax error in conditional expression: unexpected token `%'`
syntax error in conditional expression
unexpected token `%'`, expected `)`
expected `)`
unexpected argument `%'` to conditional unary operator
unexpected argument to conditional unary operator
unexpected token `%'`, conditional binary operator expected
conditional binary operator expected
unexpected argument `%'` to conditional binary operator
unexpected argument to conditional binary operator
unexpected token `%'` in conditional command
unexpected token `%'` in conditional command
unexpected token %d in conditional command
Use "%s" to leave the shell.
- CPROC
- memory exhausted
- command substitution
- newline
- select
- function
- coproc
- !"#%&'()*+,-./04
- <@JDA>F?EGHI
- ==>>>??
@@AAABBCDDDDDEEEEEEEEEEEFFFFFFFGGGGHHHHHHIIJJJKK
- OI DPWD

- ``%s'``: not a valid identifier
- `/dev/tty`
- `PROMPT_DIRTRIM`
- `getcwd`: cannot access parent directories
- syntax error: arithmetic expression required
- syntax error: ``;` unexpected
- syntax error: ``((%s))'`
- `makeheredocument`: bad instruction type %d
- here-document at line %d delimited by end-of-file (wanted ``%s'`)
- `make_redirection`: redirection instruction ``%d'` out of range
- `cleansimplecommand`
- `cprintf`: ``%c'`: invalid format character
- `<<%s%s`
- `<<< %s`
- `{%s}<&%d`
- `%d<&%d`
- `{%s}>&%d`
- `%d>&%d`
- `{%s}<&%s`
- `%d<&%s`
- `{%s}>&%s`
- `%d>&%s`
- `{%s}<&%d-`
- `%d<&%d-`
- `{%s}>&%d-`
- `%d>&%d-`
- `{%s}<&%s-`
- `%d<&%s-`
- `{%s}>&%s-`
- `%d>&%s-`
- `{%s}>&-`
- `xtrace_set`: %d: invalid file descriptor
- `xtrace_set`: NULL file pointer
- `xtrace fd (%d) != fileno xtrace fp (%d)`
- `%s=(%s)`
- `for %s in`
- `select %s in`
- `case %s in`
- `case %s in`
- `for ((`
- `print_command`: bad connector ``%d'`
- `function %s ()`
- `coproc %s`
- `print_command`
- `dispose_command`
- cannot duplicate fd %d to fd %d
- `eval_builtin`
- `builtin_env`
- `returntempenv`
- `executecondnode`
- `/dev/null`
- cannot redirect standard input from `/dev/null`: %s
- `%s_PID`
- `%s: %s: bad interpreter`
- `simple-command`
- `auto_resume`
- `substring`
- `saved_redirects`
- `%s: restricted: cannot specify `/' in command names`
- `commandnotfound_handle`
- `%s: command not found`
- `execute_coproc`: coproc [%d:%s] still exists
- `loop_redirections`
- `COLUMNS`
- `%*d%s%s`
- `execute_connection`
- `%s: readonly function`
- `execute_command`
- `%s: maximum function nesting level exceeded (%d)`
- `function_calling`
- `execute-shell-function`
- `real %2R`
- `user %2U`

- sys %2S
- TIMEFORMAT
- TIMEFORMAT: '%c': invalid format character
- pipe error
- pipe-file-descriptors
- lastpipe-exec
- execute-command
- %s: %s: cannot open as FILE
- %s: %s: invalid value for trace file descriptor
- OPTERR
- OPTIND
- ignorespace
- ignoredups
- ignoreboth
- erasedups
- HISTFILE
- all/local/variables: no function context at current scope
- make/local/variable: no function context at current scope
- shell-init
- shell level (%d) too high, resetting to 1
- popvarcontext: head of shell_variables not a function context
- popvarcontext: no global_variables context
- popscope: *head of shell/variables* not a temporary environment scope
- BASH_ARGV
- BASH_ARGC
- ~/.sh_history
- ~/.bash_history
- error importing function definition for '%s'
- /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
- HOSTTYPE
- OSTYPE
- linux-gnu
- MACHTYPE
- HOSTNAME
- MAILCHECK
- BASH_VERSION
- BASH_VERSINFO
- BASHEXECUTIONSTRING
- IGNOREEOF
- ignoreeof
- HISTCONTROL
- HISTIGNORE
- HISTTIMEFORMAT
- BASH_XTRACEFD
- SECONDS
- BASH_COMMAND
- BASH_SUBSHELL
- RANDOM
- LINENO
- BASHPID
- HISTCMD
- COMP_WORDBREAKS
- DIRSTACK
- GROUPS
- BASH_CMDS
- BASH_ALIASES
- GLOBIGNORE
- PIPESTATUS
- FUNCNEST
- HISTFILESIZE
- HISTSIZE
- HOSTFILE
- LC_ALL
- LC_COLLATE
- LC_CTYPE
- LC_MESSAGES
- LC_NUMERIC
- LC_TIME
- MAILPATH
- TERMCAP
- TERMINFO
- TEXTDOMAIN
- TEXTDOMAINDIR

- histchars
- line
- %s:%s%d:
- (null)
- last command: %s
- Aborting...
- warning:
- %s: %s:%s%d:
- %s: %s: %d
- %s: unbound variable
- %s: readonly variable
- bash-maintainers@gnu.org
- unknown command error
- bad command type
- bad connector
- bad jump
- %s%s%s: %s (error token is "%s")
- %s[%s]
- identifier expected after pre-increment or pre-decrement
- missing `)`
- syntax error: operand expected
- exponent less than 0
- division by 0
- expression expected
- `:` expected for conditional expression
- attempted assignment to non-variable
- bug: bad expassign token
- expression recursion level exceeded
- syntax error in expression
- recursion stack underflow
- invalid number
- invalid arithmetic base
- value too great for base
- */%+-&^|
- syntax error: invalid arithmetic operator
-
- Signal %d
- [%d]%c
- Stopped
- Stopped(%s)
- Running
- Done(%d)
- Exit %d
- Unknown status
- (core dumped)
- (wd: %s)
- %s: line %d:
- (core dumped)
- (wd now: %s)
- notifyofjob_status
- start_pipeline: pgrp pipe
- deleting stopped job %d with process group %ld
- [%d] %ld
- describe_pid: %ld: no such pid
- [%ld: %d (%d)] tcsetattr
- SIGCHLD trap
- waitchld: turning on WNOHANG to avoid indefinite block
- initializejobcontrol: getpgrp failed
- initializejobcontrol: setpgid
- cannot set terminal process group (%d)
- no job control in this shell
- wait_for: No record of process %ld
- waitforjob: job %d is stopped
- wait: pid %ld is not a child of this shell
- %s: job has terminated
- %s: job %d already in background
- [%d]%s
- (wd: %s)
- fork: retry
- child setpgid (%ld to %ld)
- forked pid %d appears in running job %d
- job-working-directory
- bad substitution: no closing `%' in %s

- cannot make pipe for command substitution
- cannot make child for command substitution
- command_substitute: cannot duplicate pipe as fd 1
- %s: substring expression < 0
- %s: cannot assign list to array member
- #%^,~:~=?+{ }
- #%:-=?+{ }
- %s: bad substitution
- \$%s: cannot assign in this way
- %s: parameter null or not set
- cannot make pipe for process substitution
- cannot make child for process substitution
- cannot duplicate named pipe %s as fd %d
- process substitution
- bad substitution: no closing "" in %s
- no match: %s
- You have mail in \$ _
- You have new mail in \$ _
- The mail in %s has been read
- invalid signal number
- exit trap
- debug trap
- error trap
- return trap
- interrupt trap
- runpendingtraps: bad value in trap_list[%d]: %p
- runpendingtraps: signal handler is SIG_DFL, resending %d (%s) to myself
- trap_handler: bad signal %d
- SIGHUP
- SIGINT
- SIGQUIT
- SIGILL
- SIGTRAP
- SIGABRT
- SIGBUS
- SIGFPE
- SIGKILL
- SIGUSR1
- SIGSEGV
- SIGUSR2
- SIGPIPE
- SIGALRM
- SIGTERM
- SIGSTKFLT
- SIGCHLD
- SIGCONT
- SIGSTOP
- SIGTSTP
- SIGTTIN
- SIGTTOU
- SIGURG
- SIGXCPU
- SIGXFSZ
- SIGVTALRM
- SIGPROF
- SIGWINCH
- SIGPWR
- SIGSYS
- SIGJUNK(32)
- SIGJUNK(33)
- SIGRTMIN
- SIGRTMIN+1
- SIGRTMIN+2
- SIGRTMIN+3
- SIGRTMIN+4
- SIGRTMIN+5
- SIGRTMIN+6
- SIGRTMIN+7
- SIGRTMIN+8
- SIGRTMIN+9
- SIGRTMIN+10
- SIGRTMIN+11
- SIGRTMIN+12

- SIGRTMIN+13
- SIGRTMIN+14
- SIGRTMIN+15
- SIGRTMAX-14
- SIGRTMAX-13
- SIGRTMAX-12
- SIGRTMAX-11
- SIGRTMAX-10
- SIGRTMAX-9
- SIGRTMAX-8
- SIGRTMAX-7
- SIGRTMAX-6
- SIGRTMAX-5
- SIGRTMAX-4
- SIGRTMAX-3
- SIGRTMAX-2
- SIGRTMAX-1
- SIGRTMAX
- RETURN
- cannot reset nodelay mode for fd %d
- cannot allocate new file descriptor for bash input from fd %d
- save**bash**input: buffer already exists for new fd %d
- unwind**framerun**: %s: frame not found
- unwind**framediscard**: %s: frame not found
- argument expected
- %s: integer expression expected
- %s: binary operator expected
- %s: unary operator expected
- `)' expected
- `)' expected, found %s
- missing `]'
- too many arguments
- %s.%d(%d)-%s
- release
- GNU bash, version %s (%s)
- Copyright (C) 2011 Free Software Foundation, Inc.
- License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
- This is free software; you are free to change and redistribute it.
- There is NO WARRANTY, to the extent permitted by law.
- @(#)Bash version 4.2.45(1) release GNU
- %s: cannot convert indexed to associative array
- array assign
- bad array subscript
- %s[%s: %s
- [%s]: %s
- %s: invalid associative array key
- %s: cannot assign to non-numeric index
- %s: %s: must use subscript when assigning associative array
- ;&())<>
- %s: cannot create: %s
- %s%s%s
- %s: first non-whitespace character is not ""
- no closing `%c' in %s
- \"@<>=;|&()#\$'?*![:{~
- symlink-hook
- completion-ignore-case
- ;|&{('
- \$include
- /etc/hosts
- hostname**completionfile**
- C-xC-e
- input-meta
- fc -e "\${VISUAL:-\${EDITOR:-emacs}}"
- READLINE_LINE
- READLINE_POINT
- bash**executeunix_command**
- bash**executeunix_command**: cannot find keymap for command
- fc -e vi
- fc -e "\${VISUAL:-\${EDITOR:-vi}}"
- comment-begin
- \"@<>=;|&(:
- shell-expand-line
- history-expand-line

- magic-space
- shell-forward-word
- shell-backward-word
- shell-kill-word
- shell-backward-kill-word
- alias-expand-line
- history-and-alias-expand-line
- insert-last-argument
- operate-and-get-next
- display-shell-version
- edit-and-execute-command
- complete-into-braces
- complete-filename
- possible-filename-completions
- complete-username
- possible-username-completions
- complete-hostname
- possible-hostname-completions
- complete-variable
- possible-variable-completions
- complete-command
- possible-command-completions
- glob-complete-word
- glob-expand-word
- glob-list-expansions
- dynamic-complete-history
- dabbrev-expand
- ()<>.&|
- *EmptycmD*
- %s: missing colon separator
- FIGNORE
- /usr/share/locale
- setlocale: LC_ALL: cannot change locale (%s)
- setlocale: LC_ALL: cannot change locale (%s): %s
- setlocale: %s: cannot change locale (%s)
- setlocale: %s: cannot change locale (%s): %s
- #: %s:%d
- msgid %s%s
- msgstr ""
- redirection error: cannot duplicate fd
- sh-thd
- file descriptor out of range
- %s: ambiguous redirect
- %s: cannot overwrite existing file
- %s: restricted: cannot redirect output
- cannot create temp file for here-document: %s
- %s: cannot assign fd to variable
- /dev/tcp//
- /dev/udp//
- COMP_LINE
- COMP_POINT
- COMP_TYPE
- COMP_KEY
- COMP_WORDS
- COMP_CWORD
- completion: function '%s' not found
- gen-shell-function-matches
- COMPREPLY
- *DefaultCmD*
- programmable_completion: %s: possible retry loop
- progcomp_insert: %s: NULL COMPSPEC
- %s: cannot allocate %lu bytes (%lu bytes allocated)
- xmalloc
- xrealloc
- Read lines from a file into an array variable.
- A synonym for `mapfile'.
- Read lines from the standard input into an indexed array variable.
- Read lines from the standard input into the indexed array variable ARRAY, or from file descriptor FD if the -u option is supplied. The variable MAPFILE
- is the default ARRAY.
- Options:
- -n count
- Copy at most COUNT lines. If COUNT is 0, all lines are copied.

- -O origin
- Begin assigning to ARRAY at index ORIGIN. The default index is 0.
- -s count
- Discard the first COUNT lines read.
- -t
- Remove a trailing newline from each line read.
- -u fd
- Read lines from file descriptor FD instead of the standard input.
- -C callback
- Evaluate CALLBACK each time QUANTUM lines are read.
- -c quantum
- Specify the number of lines read between each call to CALLBACK.
- Arguments:
- ARRAY
- Array variable name to use for file data.
- If -C is supplied without -c, the default quantum is 5000. When
- CALLBACK is evaluated, it is supplied the index of the next array
- element to be assigned and the line to be assigned to that element
- as additional arguments.
- If not supplied with an explicit origin, mapfile will clear ARRAY before
- assigning to it.
- Exit Status:
- Returns success unless an invalid option is given or ARRAY is readonly or
- not an indexed array.
- Modify or display completion options.
- Modify the completion options for each NAME, or, if no NAMES are supplied,
- the completion currently being executed. If no OPTIONS are given, print
- the completion options for each NAME or the current completion specification.
- Options:
- -o option
- Set completion option OPTION for each NAME
- Change options for the "default" command completion
- Change options for the "empty" command completion
- Using `+o' instead of -o'` turns off the specified option.
- Arguments:
- Each NAME refers to a command for which a completion specification must
- have previously been defined using the `'complete'` builtin. If no NAMES
- are supplied, compopt must be called by a function currently generating
- completions, and the options for that currently-executing completion
- generator are modified.
- Exit Status:
- Returns success unless an invalid option is supplied or NAME does not
- have a completion specification defined.
- Display possible completions depending on the options.
- Intended to be used from within a shell function generating possible
- completions. If the optional WORD argument is supplied, matches against
- WORD are generated.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Specify how arguments are to be completed by Readline.
- For each NAME, specify how arguments are to be completed. If no options
- are supplied, existing completion specifications are printed in a way that
- allows them to be reused as input.
- Options:
- -p
- print existing completion specifications in a reusable format
- -r
- remove a completion specification for each NAME, or, if no
- NAMES are supplied, all completion specifications
- -D
- apply the completions and actions as the default for commands
- without any specific completion defined
- -E
- apply the completions and actions to "empty" commands --
- completion attempted on a blank line
- When completion is attempted, the actions are applied in the order the
- uppercase-letter options are listed above. The -D option takes
- precedence over -E.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Formats and prints ARGUMENTS under control of the FORMAT.
- Options:
- -v var

- assign the output to shell variable VAR rather than
- display it on the standard output
- FORMAT is a character string which contains three types of objects: plain
- characters, which are simply copied to standard output; character escape
- sequences, which are converted and copied to the standard output; and
- format specifications, each of which causes printing of the next successive
- argument.
- In addition to the standard format specifications described in printf(1)
- and printf(3), printf interprets:
- %b
- expand backslash escape sequences in the corresponding argument
- %q
- quote the argument in a way that can be reused as shell input
- %(fmt)T output the date-time string resulting from using FMT as a format
- string for strftime(3)
- Exit Status:
- Returns success unless an invalid option is given or a write or assignment
- error occurs.
- Set and unset shell options.
- Change the setting of each shell option OPTNAME. Without any option
- arguments, list all shell options with an indication of whether or not each
- is set.
- Options:
- -o
- restrict OPTNAMEs to those defined for use with 'set -o'
- -p
- print each shell option with an indication of its status
- -q
- suppress output
- -s
- enable (set) each OPTNAME
- -u
- disable (unset) each OPTNAME
- Exit Status:
- Returns success if OPTNAME is enabled; fails if an invalid option is
- given or OPTNAME is disabled.
- Display directory stack.
- Display the list of currently remembered directories. Directories
- find their way onto the list with the 'pushd' command; you can get
- back up through the list with the 'popd' command.
- Options:
- -c
- clear the directory stack by deleting all of the elements
- -l
- do not print tilde-prefixed versions of directories relative
- to your home directory
- -p
- print the directory stack with one entry per line
- -v
- print the directory stack with one entry per line prefixed
- with its position in the stack
- Arguments:
- +N
- Displays the Nth entry counting from the left of the list shown by
- dirs when invoked without options, starting with zero.
- -N
- Displays the Nth entry counting from the right of the list shown by
- dirs when invoked without options, starting with zero.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Remove directories from stack.
- Removes entries from the directory stack. With no arguments, removes
- the top directory from the stack, and changes to the new top directory.
- Options:
- -n
- Suppresses the normal change of directory when removing
- directories from the stack, so only the stack is manipulated.
- Arguments:
- +N
- Removes the Nth entry counting from the left of the list
- shown by 'dirs', starting with zero. For example: popd +0'
- removes the first directory, 'popd +1' the second.
- -N

- Removes the Nth entry counting from the right of the list
- shown by `dirs'`, starting with zero. For example: `popd -0'`
- removes the last directory, `'popd -1'` the next to last.
- The `'dirs'` builtin displays the directory stack.
- Exit Status:
- Returns success unless an invalid argument is supplied or the directory
- change fails.
- Add directories to stack.
- Adds a directory to the top of the directory stack, or rotates
- the stack, making the new top of the stack the current working
- directory. With no arguments, exchanges the top two directories.
- Options:
- `-n`
- Suppresses the normal change of directory when adding
- directories to the stack, so only the stack is manipulated.
- Arguments:
- `+N`
- Rotates the stack so that the Nth directory (counting
- from the left of the list shown by `'dirs'`, starting with
- zero) is at the top.
- `-N`
- Rotates the stack so that the Nth directory (counting
- from the right of the list shown by `'dirs'`, starting with
- zero) is at the top.
- `dir`
- Adds DIR to the directory stack at the top, making it the
- new current working directory.
- The `'dirs'` builtin displays the directory stack.
- Exit Status:
- Returns success unless an invalid argument is supplied or the directory
- change fails.
- Common shell variable names and usage.
- BASH_VERSION
- Version information for this Bash.
- CDPATH
- A colon-separated list of directories to search
- for directories given as arguments to `'cd'`.
- GLOBIGNORE
- A colon-separated list of patterns describing filenames to
- be ignored by pathname expansion.
- HISTFILE
- The name of the file where your command history is stored.
- HISTFILESIZE
- The maximum number of lines this file can contain.
- HISTSIZE
- The maximum number of history lines that a running
- shell can access.
- HOME
- The complete pathname to your login directory.
- HOSTNAME
- The name of the current host.
- HOSTTYPE
- The type of CPU this version of Bash is running under.
- IGNOREEOF
- Controls the action of the shell on receipt of an EOF
- character as the sole input. If set, then the value
- of it is the number of EOF characters that can be seen
- in a row on an empty line before the shell will exit
- (default 10). When unset, EOF signifies the end of input.
- MACHTYPE
- A string describing the current system Bash is running on.
- MAILCHECK
- How often, in seconds, Bash checks for new mail.
- MAILPATH
- A colon-separated list of filenames which Bash checks
- for new mail.
- OSTYPE
- The version of Unix this version of Bash is running on.
- PATH
- A colon-separated list of directories to search when
- looking for commands.
- PROMPT_COMMAND
- A command to be executed before the printing of each

- primary prompt.
- PS1
- The primary prompt string.
- PS2
- The secondary prompt string.
- PWD
- The full pathname of the current directory.
- SHELLOPTS
- A colon-separated list of enabled shell options.
- TERM
- The name of the current terminal type.
- TIMEFORMAT
- The output format for timing statistics displayed by the
- 'time' reserved word.
- auto_resume
- Non-null means a command word appearing on a line by
- itself is first looked for in the list of currently
- stopped jobs. If found there, that job is foregrounded.
- A value of 'exact' means that the command word must
- exactly match a command in the list of stopped jobs. A
- value of 'substring' means that the command word must
- match a substring of the job. Any other value means that
- the command must be a prefix of a stopped job.
- histchars
- Characters controlling history expansion and quick
- substitution. The first character is the history
- substitution character, usually '!'. The second is
- the quick substitution character, usually '^'. The
- third is the history comment character, usually '#'. The
- HISTIGNORE
- A colon-separated list of patterns used to decide which
- commands should be saved on the history list.
- Execute conditional command.
- Returns a status of 0 or 1 depending on the evaluation of the conditional
- expression EXPRESSION. Expressions are composed of the same primaries used
- by the 'test' builtin, and may be combined using the following operators:
- (EXPRESSION)
- Returns the value of EXPRESSION
- ! EXPRESSION
- True if EXPRESSION is false; else false
- EXPR1 && EXPR2
- True if both EXPR1 and EXPR2 are true; else false
- EXPR1 || EXPR2
- True if either EXPR1 or EXPR2 is true; else false
- When the '==' and '!=' operators are used, the string to the right of
- the operator is used as a pattern and pattern matching is performed.
- When the '=~' operator is used, the string to the right of the operator
- is matched as a regular expression.
- The && and || operators do not evaluate EXPR2 if EXPR1 is sufficient to
- determine the expression's value.
- Exit Status:
- 0 or 1 depending on value of EXPRESSION.
- Evaluate arithmetic expression.
- The EXPRESSION is evaluated according to the rules for arithmetic
- evaluation. Equivalent to "let EXPRESSION".
- Exit Status:
- Returns 1 if EXPRESSION evaluates to 0; returns 0 otherwise.
- Resume job in foreground.
- Equivalent to the JOB_SPEC argument to the 'fg' command. Resume a
- stopped or background job. JOB_SPEC can specify either a job name
- or a job number. Following JOB_SPEC with a '&' places the job in
- the background, as if the job specification had been supplied as an
- argument to 'bg'.
- Exit Status:
- Returns the status of the resumed job.
- Group commands as a unit.
- Run a set of commands in a group. This is one way to redirect an
- entire set of commands.
- Exit Status:
- Returns the status of the last command executed.
- Define shell function.
- Create a shell function named NAME. When invoked as a simple command,
- NAME runs COMMANDs in the calling shell's context. When NAME is invoked,

- the arguments are passed to the function as \$1...\$n, and the function's
- name is in \$FUNCNAME.
- Exit Status:
- Returns success unless NAME is readonly.
- Create a coprocess named NAME.
- Execute COMMAND asynchronously, with the standard output and standard
- input of the command connected via a pipe to file descriptors assigned
- to indices 0 and 1 of an array variable NAME in the executing shell.
- The default NAME is "COPROC".
- Exit Status:
- Returns the exit status of COMMAND.
- Execute commands as long as a test does not succeed.
- Expand and execute COMMANDS as long as the final command in the
- 'until' COMMANDS has an exit status which is not zero.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands as long as a test succeeds.
- Expand and execute COMMANDS as long as the final command in the
- 'while' COMMANDS has an exit status of zero.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands based on conditional.
- The 'if COMMANDS' list is executed. If its exit status is zero, then the
- then COMMANDS' list is executed. Otherwise, each elif COMMANDS' list is
- executed in turn, and if its exit status is zero, the corresponding
- 'then COMMANDS' list is executed and the if command completes. Otherwise,
- the 'else COMMANDS' list is executed, if present. The exit status of the
- entire construct is the exit status of the last command executed, or zero
- if no condition tested true.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands based on pattern matching.
- Selectively execute COMMANDS based upon WORD matching PATTERN. The
- '[' is used to separate multiple patterns.
- Exit Status:
- Returns the status of the last command executed.
- Report time consumed by pipeline's execution.
- Execute PIPELINE and print a summary of the real time, user CPU time,
- and system CPU time spent executing PIPELINE when it terminates.
- Options:
- -p
- print the timing summary in the portable Posix format
- The value of the TIMEFORMAT variable is used as the output format.
- Exit Status:
- The return status is the return status of PIPELINE.
- Select words from a list and execute commands.
- The WORDS are expanded, generating a list of words. The
- set of expanded words is printed on the standard error, each
- preceded by a number. If in WORDS' is not present, in "\$@"
- is assumed. The PS3 prompt is then displayed and a line read
- from the standard input. If the line consists of the number
- corresponding to one of the displayed words, then NAME is set
- to that word. If the line is empty, WORDS and the prompt are
- redisplayed. If EOF is read, the command completes. Any other
- value read causes NAME to be set to null. The line read is saved
- in the variable REPLY. COMMANDS are executed after each selection
- until a break command is executed.
- Exit Status:
- Returns the status of the last command executed.
- Arithmetic for loop.
- Equivalent to
- ((EXP1))
- while ((EXP2)); do
- COMMANDS
- ((EXP3))
- EXP1, EXP2, and EXP3 are arithmetic expressions. If any expression is
- omitted, it behaves as if it evaluates to 1.
- Exit Status:
- Returns the status of the last command executed.
- Execute commands for each member in a list.
- The 'for' loop executes a sequence of commands for each member in a
- list of items. If in WORDS ...;' is not present, then in "\$@" is
- assumed. For each element in WORDS, NAME is set to that element, and

- the COMMANDS are executed.
- Exit Status:
- Returns the status of the last command executed.
- Wait for job completion and return exit status.
- Waits for the process identified by ID, which may be a process ID or a job specification, and reports its termination status. If ID is not given, waits for all currently active child processes, and the return status is zero. If ID is a job specification, waits for all processes in the job's pipeline.
- Exit Status:
- Returns the status of ID; fails if ID is invalid or an invalid option is given.
- Display or set file mode mask.
- Sets the user file-creation mask to MODE. If MODE is omitted, prints the current value of the mask.
- If MODE begins with a digit, it is interpreted as an octal number; otherwise it is a symbolic mode string like that accepted by chmod(1).
- Options:
- -p
- if MODE is omitted, output in a form that may be reused as input
- -S
- makes the output symbolic; otherwise an octal number is output
- Exit Status:
- Returns success unless MODE is invalid or an invalid option is given.
- Modify shell resource limits.
- Provides control over the resources available to the shell and processes it creates, on systems that allow such control.
- Options:
- -S
- use the 'soft' resource limit
- -H
- use the 'hard' resource limit
- -a
- all current limits are reported
- -b
- the socket buffer size
- -c
- the maximum size of core files created
- -d
- the maximum size of a process's data segment
- -e
- the maximum scheduling priority ('nice')
- -f
- the maximum size of files written by the shell and its children
- -i
- the maximum number of pending signals
- -l
- the maximum size a process may lock into memory
- -m
- the maximum resident set size
- -n
- the maximum number of open file descriptors
- -p
- the pipe buffer size
- -q
- the maximum number of bytes in POSIX message queues
- -r
- the maximum real-time scheduling priority
- -s
- the maximum stack size
- -t
- the maximum amount of cpu time in seconds
- -u
- the maximum number of user processes
- -v
- the size of virtual memory
- -x
- the maximum number of file locks
- If LIMIT is given, it is the new value of the specified resource; the special LIMIT values 'soft', 'hard', and 'unlimited' stand for the current soft limit, the current hard limit, and no limit, respectively.
- Otherwise, the current value of the specified resource is printed. If no option is given, then -f is assumed.

- Values are in 1024-byte increments, except for -t, which is in seconds,
- -p, which is in increments of 512 bytes, and -u, which is an unscaled
- number of processes.
- Exit Status:
- Returns success unless an invalid option is supplied or an error occurs.
- Display information about command type.
- For each NAME, indicate how it would be interpreted if used as a
- command name.
- Options:
- -a
- display all locations containing an executable named NAME;
- includes aliases, builtins, and functions, if and only if
- the -p option is not also used
- -f
- suppress shell function lookup
- -P
- force a PATH search for each NAME, even if it is an alias,
- builtin, or function, and returns the name of the disk file
- that would be executed
- -p
- returns either the name of the disk file that would be executed,
- or nothing if `type -t NAME` would not return file'.
- -t
- output a single word which is one of `alias`, `keyword`,
- `function`, `builtin`, `file` or `'`, if NAME is an alias, shell
- reserved word, shell function, shell builtin, disk file, or not
- found, respectively
- Arguments:
- NAME
- Command name to be interpreted.
- Exit Status:
- Returns success if all of the NAMEs are found; fails if any are not found.
- Trap signals and other events.
- Defines and activates handlers to be run when the shell receives signals
- or other conditions.
- ARG is a command to be read and executed when the shell receives the
- signal(s) SIGNALSPEC. If ARG is absent (and a single SIGNALSPEC
- is supplied) or `'`, each specified signal is reset to its original
- value. If ARG is the null string each SIGNAL_SPEC is ignored by the
- shell and by the commands it invokes.
- If a SIGNAL_SPEC is EXIT (0) ARG is executed on exit from the shell. If
- a SIGNAL_SPEC is DEBUG, ARG is executed before every simple command. If
- a SIGNAL_SPEC is RETURN, ARG is executed each time a shell function or a
- script run by the `.` or source builtins finishes executing. A SIGNAL_SPEC
- of ERR means to execute ARG each time a command's failure would cause the
- shell to exit when the -e option is enabled.
- If no arguments are supplied, trap prints the list of commands associated
- with each signal.
- Options:
- -l
- print a list of signal names and their corresponding numbers
- -p
- display the trap commands associated with each SIGNAL_SPEC
- Each SIGNAL_SPEC is either a signal name in or a signal number.
- Signal names are case insensitive and the SIG prefix is optional. A
- signal may be sent to the shell with `"kill -signal $$"`.
- Exit Status:
- Returns success unless a SIGSPEC is invalid or an invalid option is given.
- Display process times.
- Prints the accumulated user and system times for the shell and all of its
- child processes.
- Exit Status:
- Always succeeds.
- Evaluate conditional expression.
- This is a synonym for the "test" builtin, but the last argument must
- be a literal `]`, to match the opening `[`.
- Evaluate conditional expression.
- Exits with a status of 0 (true) or 1 (false) depending on
- the evaluation of EXPR. Expressions may be unary or binary. Unary
- expressions are often used to examine the status of a file. There
- are string operators and numeric comparison operators as well.
- The behavior of test depends on the number of arguments. Read the
- bash manual page for the complete specification.

- File operators:
- -a FILE True if file exists.
- -b FILE True if file is block special.
- -c FILE True if file is character special.
- -d FILE True if file is a directory.
- -e FILE True if file exists.
- -f FILE True if file exists and is a regular file.
- -g FILE True if file is set-group-id.
- -h FILE True if file is a symbolic link.
- -L FILE True if file is a symbolic link.
- -k FILE True if file has its 'sticky' bit set.
- -p FILE True if file is a named pipe.
- -r FILE True if file is readable by you.
- -s FILE True if file exists and is not empty.
- -S FILE True if file is a socket.
- -t FD True if FD is opened on a terminal.
- -u FILE True if the file is set-user-id.
- -w FILE True if the file is writable by you.
- -x FILE True if the file is executable by you.
- -O FILE True if the file is effectively owned by you.
- -G FILE True if the file is effectively owned by your group.
- -N FILE True if the file has been modified since it was last read.
- FILE1 -nt FILE2 True if file1 is newer than file2 (according to modification date).
- FILE1 -ot FILE2 True if file1 is older than file2.
- FILE1 -ef FILE2 True if file1 is a hard link to file2.
- String operators:
- -z STRING True if string is empty.
- -n STRING
- STRING True if string is not empty.
- STRING1 = STRING2
- True if the strings are equal.
- STRING1 != STRING2
- True if the strings are not equal.
- STRING1 < STRING2
- True if STRING1 sorts before STRING2 lexicographically.
- STRING1 > STRING2
- True if STRING1 sorts after STRING2 lexicographically.
- Other operators:
- -o OPTION True if the shell option OPTION is enabled.
- -v VAR
- True if the shell variable VAR is set
- ! EXPR True if expr is false.
- EXPR1 -a EXPR2 True if both expr1 AND expr2 are true.
- EXPR1 -o EXPR2 True if either expr1 OR expr2 is true.
- arg1 OP arg2 Arithmetic tests. OP is one of -eq, -ne,
- -lt, -le, -gt, or -ge.
- Arithmetic binary operators return true if ARG1 is equal, not-equal,
- less-than, less-than-or-equal, greater-than, or greater-than-or-equal
- than ARG2.
- Exit Status:
- Returns success if EXPR evaluates to true; fails if EXPR evaluates to false or an invalid argument is given.
- Suspend shell execution.
- Suspend the execution of this shell until it receives a SIGCONT signal.
- Unless forced, login shells cannot be suspended.
- Options:
- -f
- force the suspend, even if the shell is a login shell
- Exit Status:
- Returns success unless job control is not enabled or an error occurs.
- Execute commands from a file in the current shell.
- Read and execute commands from FILENAME in the current shell. The
- entries in \$PATH are used to find the directory containing FILENAME.
- If any ARGUMENTS are supplied, they become the positional parameters
- when FILENAME is executed.
- Exit Status:
- Returns the status of the last command executed in FILENAME; fails if
- FILENAME cannot be read.
- Shift positional parameters.
- Rename the positional parameters \$N+1,\$N+2 ... to \$1,\$2 ... If N is
- not given, it is assumed to be 1.
- Exit Status:

- Returns success unless N is negative or greater than \$#.
- Mark shell variables as unchangeable.
- Mark each NAME as read-only; the values of these NAMEs may not be changed by subsequent assignment. If VALUE is supplied, assign VALUE before marking as read-only.
- Options:
 - -a
 - refer to indexed array variables
 - -A
 - refer to associative array variables
 - -f
 - refer to shell functions
 - -p
 - display a list of all readonly variables and functions
- An argument of '--' disables further option processing.
- Exit Status:
 - Returns success unless an invalid option is given or NAME is invalid.
 - Set export attribute for shell variables.
 - Marks each NAME for automatic export to the environment of subsequently executed commands. If VALUE is supplied, assign VALUE before exporting.
- Options:
 - -f
 - refer to shell functions
 - -n
 - remove the export property from each NAME
 - -p
 - display a list of all exported variables and functions
- An argument of '--' disables further option processing.
- Exit Status:
 - Returns success unless an invalid option is given or NAME is invalid.
 - Unset values and attributes of shell variables and functions.
 - For each NAME, remove the corresponding variable or function.
- Options:
 - -f
 - treat each NAME as a shell function
 - -v
 - treat each NAME as a shell variable
- Without options, unset first tries to unset a variable, and if that fails, tries to unset a function.
- Some variables cannot be unset; also see `readonly`.
- Exit Status:
 - Returns success unless an invalid option is given or a NAME is read-only.
 - Set or unset values of shell options and positional parameters.
 - Change the value of shell attributes and positional parameters, or display the names and values of shell variables.
- Options:
 - -a Mark variables which are modified or created for export.
 - -b Notify of job termination immediately.
 - -e Exit immediately if a command exits with a non-zero status.
 - -f Disable file name generation (globbing).
 - -h Remember the location of commands as they are looked up.
 - -k All assignment arguments are placed in the environment for a command, not just those that precede the command name.
 - -m Job control is enabled.
 - -n Read commands but do not execute them.
 - -o option-name
 - Set the variable corresponding to option-name:
 - allexport same as -a
 - braceexpand same as -B
 - emacs use an emacs-style line editing interface
 - errexit same as -e
 - errtrace same as -E
 - functrace same as -T
 - hashall same as -h
 - histexpand same as -H
 - history enable command history
 - ignoreeof the shell will not exit upon reading EOF
 - interactive-comments
 - allow comments to appear in interactive commands
 - keyword same as -k
 - monitor same as -m
 - noclobber same as -C
 - noexec same as -n

- noglob same as -f
- nolog currently accepted but ignored
- notify same as -b
- nounset same as -u
- onecmd same as -t
- physical same as -P
- pipefail the return value of a pipeline is the status of
- the last command to exit with a non-zero status,
- or zero if no command exited with a non-zero status
- posix change the behavior of bash where the default
- operation differs from the Posix standard to
- match the standard
- privileged same as -p
- verbose same as -v
- vi use a vi-style line editing interface
- xtrace same as -x
- -p Turned on whenever the real and effective user ids do not match.
- Disables processing of the \$ENV file and importing of shell
- functions. Turning this option off causes the effective uid and
- gid to be set to the real uid and gid.
- -t Exit after reading and executing one command.
- -u Treat unset variables as an error when substituting.
- -v Print shell input lines as they are read.
- -x Print commands and their arguments as they are executed.
- -B the shell will perform brace expansion
- -C If set, disallow existing regular files to be overwritten
- by redirection of output.
- -E If set, the ERR trap is inherited by shell functions.
- -H Enable ! style history substitution. This flag is on
- by default when the shell is interactive.
- -P If set, do not follow symbolic links when executing commands
- such as cd which change the current directory.
- -T If set, the DEBUG trap is inherited by shell functions.
- -- Assign any remaining arguments to the positional parameters.
- If there are no remaining arguments, the positional parameters
- are unset.
-
- Assign any remaining arguments to the positional parameters.
- The -x and -v options are turned off.
- Using + rather than - causes these flags to be turned off. The
- flags can also be used upon invocation of the shell. The current
- set of flags may be found in \$-. The remaining n ARGs are positional
- parameters and are assigned, in order, to \$1, \$2, ... \$n. If no
- ARGs are given, all shell variables are printed.
- Exit Status:
- Returns success unless an invalid option is given.
- Return from a shell function.
- Causes a function or sourced script to exit with the return value
- specified by N. If N is omitted, the return status is that of the
- last command executed within the function or script.
- Exit Status:
- Returns N, or failure if the shell is not executing a function or script.
- Read a line from the standard input and split it into fields.
- Reads a single line from the standard input, or from file descriptor FD
- if the -u option is supplied. The line is split into fields as with word
- splitting, and the first word is assigned to the first NAME, the second
- word to the second NAME, and so on, with any leftover words assigned to
- the last NAME. Only the characters found in \$IFS are recognized as word
- delimiters.
- If no NAMEs are supplied, the line read is stored in the REPLY variable.
- Options:
- -a array
- assign the words read to sequential indices of the array
- variable ARRAY, starting at zero
- -d delim
- continue until the first character of DELIM is read, rather
- than newline
- -e
- use Readline to obtain the line in an interactive shell
- -i text
- Use TEXT as the initial text for Readline
- -n nchars
- return after reading NCHARS characters rather than waiting

- for a newline, but honor a delimiter if fewer than NCHARS
- characters are read before the delimiter
- -N nchars
- return only after reading exactly NCHARS characters, unless
- EOF is encountered or read times out, ignoring any delimiter
- -p prompt
- output the string PROMPT without a trailing newline before
- attempting to read
- -r
- do not allow backslashes to escape any characters
- -s
- do not echo input coming from a terminal
- -t timeout
- time out and return failure if a complete line of input is
- not read within TIMEOUT seconds. The value of the TIMEOUT
- variable is the default timeout. TIMEOUT may be a
- fractional number. If TIMEOUT is 0, read returns success only
- if input is available on the specified file descriptor. The
- exit status is greater than 128 if the timeout is exceeded
- -u fd
- read from file descriptor FD instead of the standard input
- Exit Status:
- The return code is zero, unless end-of-file is encountered, read times out,
- or an invalid file descriptor is supplied as the argument to -u.
- Evaluate arithmetic expressions.
- Evaluate each ARG as an arithmetic expression. Evaluation is done in
- fixed-width integers with no check for overflow, though division by 0
- is trapped and flagged as an error. The following list of operators is
- grouped into levels of equal-precedence operators. The levels are listed
- in order of decreasing precedence.
- id++, id--
- variable post-increment, post-decrement
- ++id, --id
- variable pre-increment, pre-decrement
- unary minus, plus
- logical and bitwise negation
- exponentiation
- ., /, %
- *multiplication, division, remainder*
- *addition, subtraction*
- <<, >>
- *left and right bitwise shifts*
- <=, >=, <, >
- *comparison*
- ==, !=
- *equality, inequality*
- *bitwise AND*
- *bitwise XOR*
- *bitwise OR*
- *logical AND*
- *logical OR*
- *expr ? expr : expr*
- *conditional operator*
- =, *=, /=, %=
- +=, -=, <<=, >>=
- &=, ^=, |=
- *assignment*
- *Shell variables are allowed as operands. The name of the variable*
- *is replaced by its value (coerced to a fixed-width integer) within*
- *an expression. The variable need not have its integer attribute*
- *turned on to be used in an expression.*
- *Operators are evaluated in order of precedence. Sub-expressions in*
- *parentheses are evaluated first and may override the precedence*
- *rules above.*
- Exit Status:
- *If the last ARG evaluates to 0, let returns 1; let returns 0 otherwise.*
- *Send a signal to a job.*
- *Send the processes identified by PID or JOBSPEC the signal named by*
- *SIGSPEC or SIGNUM. If neither SIGSPEC nor SIGNUM is present, then*
- *SIGTERM is assumed.*
- Options:
- -s sig
- SIG is a signal name

- `-n sig`
- *SIG is a signal number*
- `-l`
- *list the signal names; if arguments follow -l' they are* `<li>assumed to be signal numbers for which names should be listed</li>` `<li>Kill is a shell builtin for two reasons: it allows job IDs to be used</li>` `<li>instead of process IDs, and allows processes to be killed if the limit</li>` `<li>on processes that you can create is reached.</li>` `<li>Exit Status:</li>` `<li>Returns success unless an invalid option is given or an error occurs.</li>` `<li>Remove jobs from current shell.</li>` `<li>Removes each JOBSPEC argument from the table of active jobs. Without</li>` `<li>any JOBSPECs, the shell uses its notion of the current job.</li>` `<li>Options:</li>` `<li>-a</li>` `<li>remove all jobs if JOBSPEC is not supplied</li>` `<li>-h</li>` `<li>mark each JOBSPEC so that SIGHUP is not sent to the job if the</li>` `<li>shell receives a SIGHUP</li>` `<li>-r</li>` `<li>remove only running jobs</li>` `<li>Exit Status:</li>` `<li>Returns success unless an invalid option or JOBSPEC is given.</li>` `<li>Display status of jobs.</li>` `<li>Lists the active jobs. JOBSPEC restricts output to that job.</li>` `<li>Without options, the status of all active jobs is displayed.</li>` `<li>Options:</li>` `<li>-l</li>` `<li>lists process IDs in addition to the normal information</li>` `<li>-n</li>` `<li>list only processes that have changed status since the last</li>` `<li>notification</li>` `<li>-p</li>` `<li>lists process IDs only</li>` `<li>-r</li>` `<li>restrict output to running jobs</li>` `<li>-s</li>` `<li>restrict output to stopped jobs</li>` `<li>If -x is supplied, COMMAND is run after all job specifications that</li>` `<li>appear in ARGS have been replaced with the process ID of that job's</li>` `<li>process group leader.</li>` `<li>Exit Status:</li>` `<li>Returns success unless an invalid option is given or an error occurs.</li>` `<li>If -x is used, returns the exit status of COMMAND.</li>` `<li>Display or manipulate the history list.</li>` `<li>Display the history list with line numbers, prefixing each modified</li>` `<li>entry with a ' . An argument of N lists only the last N entries.`
- Options:
 - `-c`
 - clear the history list by deleting all of the entries
 - `-d offset`
 - delete the history entry at offset OFFSET.
 - `-a`
 - append history lines from this session to the history file
 - `-n`
 - read all history lines not already read from the history file
 - `-r`
 - read the history file and append the contents to the history
 - `-w`
 - write the current history to the history file
 - and append them to the history list
 - `-p`
 - perform history expansion on each ARG and display the result
 - without storing it in the history list
 - `-s`
 - append the ARGs to the history list as a single entry
 - If FILENAME is given, it is used as the history file. Otherwise,
 - if \$HISTFILE has a value, that is used, else ~/.bashhistory.
 - If the \$HISTTIMEFORMAT variable is set and not null, its value is used
 - as a format string for strftime(3) to print the time stamp associated
 - with each displayed history entry. No time stamps are printed otherwise.
 - Exit Status:
 - Returns success unless an invalid option is given or an error occurs.
 - Display information about builtin commands.
 - Displays brief summaries of builtin commands. If PATTERN is
 - specified, gives detailed help on all commands matching PATTERN,
 - otherwise the list of help topics is printed.
 - Options:
 - `-d`
 - output short description for each topic
 - `-m`
 - display usage in pseudo-manpage format
 - `-s`
 - output only a short usage synopsis for each topic matching
 - PATTERN
 - Arguments:
 - PATTERN
 - Pattern specifying a help topic
 - Exit Status:
 - Returns success unless PATTERN is not found or an invalid option is given.
 - Remember or display program locations.
 - Determine and remember the full pathname of each command NAME. If
 - no arguments are given, information about remembered commands is displayed.
 - Options:
 - `-d`
 - forget the remembered location of each NAME
 - `-l`
 - display in a format that may be reused as input
 - `-p pathname`
 - use PATHNAME is the full pathname of NAME
 - `-r`
 - forget all remembered locations
 - `-t`

- *print the remembered location of each NAME, preceding*
- *each location with the corresponding NAME if multiple*
- *NAMEs are given*
- *Arguments:*
- *NAME*
- *Each NAME is searched for in \$PATH and added to the list*
- *of remembered commands.*
- *Exit Status:*
- *Returns success unless NAME is not found or an invalid option is given.*
- *Move jobs to the background.*
- *Place the jobs identified by each JOBSPEC in the background, as if they*
- *had been started with* `&name;'. If JOB_SPEC is not present, the shell's notion of the current job is used.` `Exit Status:` `Returns success unless job control is not enabled or an error occurs.` `Move job to the foreground.` `Place the job identified by JOB_SPEC in the foreground, making it the current job. If JOB_SPEC is not present, the shell's notion of the current job is used.` `Exit Status:` `Status of command placed in foreground, or failure if an error occurs.` `Display or execute commands from the history list.` `fc is used to list or edit and re-execute commands from the history list.` `FIRST and LAST can be numbers specifying the range, or FIRST can be a string, which means the most recent command beginning with that string.` `Options:` `-e ENAME` `select which editor to use. Default is FCEDIT, then EDITOR,` `then vi` `-l` `list lines instead of editing` `-n` `omit line numbers when listing` `-r` `reverse the order of the lines (newest listed first)` `With the fc -s [pat=rep ...]` `[command]` format, COMMAND is
- *re-executed after the substitution OLD=NEW is performed.*
- *A useful alias to use with this is r=fc -s', so that typing* `r cc'` `runs the last command beginning with` `cc'` `and typing` `r'` `re-executes` `the last command.` `Exit Status:` `Returns success or status of executed command; non-zero if an error occurs.` `Exit a login shell.` `Exits a login shell with exit status N. Returns an error if not executed in a login shell.` `Exit the shell.` `Exits the shell with a status of N. If N is omitted, the exit status is that of the last command executed.` `Replace the shell with the given command.` `Execute COMMAND, replacing this shell with the specified program.` `ARGUMENTS become the arguments to COMMAND. If COMMAND is not specified,` `any redirections take effect in the current shell.` `Options:` `-a name` `pass NAME as the zeroth argument to COMMAND` `-c` `execute COMMAND with an empty environment` `-l` `place a dash in the zeroth argument to COMMAND` `If the command cannot be executed, a non-interactive shell exits, unless the shell option` **execfail** `is set.`
- *Exit Status:*
- *Returns success unless COMMAND is not found or a redirection error occurs.*
- *Parse option arguments.*
- *Getopts is used by shell procedures to parse positional parameters*
- *as options.*
- *OPTSTRING contains the option letters to be recognized; if a letter*
- *is followed by a colon, the option is expected to have an argument,*
- *which should be separated from it by white space.*
- *Each time it is invoked, getopts will place the next option in the*
- *shell variable \$name, initializing name if it does not exist, and*
- *the index of the next argument to be processed into the shell*
- *variable OPTIND. OPTIND is initialized to 1 each time the shell or*
- *a shell script is invoked. When an option requires an argument,*
- *getopts places that argument into the shell variable OPTARG.*
- *getopts reports errors in one of two ways. If the first character*
- *of OPTSTRING is a colon, getopts uses silent error reporting. In*
- *this mode, no error messages are printed. If an invalid option is*
- *seen, getopts places the option character found into OPTARG. If a*
- *required argument is not found, getopts places a ':' into NAME and*
- *sets OPTARG to the option character found. If getopts is not in*
- *silent mode, and an invalid option is seen, getopts places '?' into*
- *NAME and unsets OPTARG. If a required argument is not found, a '?'*
- *is placed in NAME, OPTARG is unset, and a diagnostic message is*
- *printed.*
- *If the shell variable OPTERR has the value 0, getopts disables the*
- *printing of error messages, even if the first character of*
- *OPTSTRING is not a colon. OPTERR has the value 1 by default.*
- *Getopts normally parses the positional parameters (\$0 - \$9), but if*
- *more arguments are given, they are parsed instead.*
- *Exit Status:*
- *Returns success if an option is found; fails if the end of options is*
- *encountered or an error occurs.*
- *Execute arguments as a shell command.*
- *Combine ARGs into a single string, use the result as input to the shell,*
- *and execute the resulting commands.*
- *Exit Status:*
- *Returns exit status of command or success if command is null.*
- *Enable and disable shell builtins.*
- *Enables and disables builtin shell commands. Disabling allows you to*
- *execute a disk command which has the same name as a shell builtin*
- *without using a full pathname.*
- *Options:*
- *-a*
- *print a list of builtins showing whether or not each is enabled*

- -n
- disable each NAME or display a list of disabled builtins
- -p
- print the list of builtins in a reusable format
- -s
- **print only the names of Posix special' builtins** Options controlling dynamic loading: -f Load builtin NAME from shared object FILENAME -d Remove a builtin loaded with -f Without options, each NAME is enabled. To use the **test'** found in \$PATH instead of the shell builtin
- **version, type** enable -n test'. Exit Status: Returns success unless NAME is not a shell builtin or an error occurs. Write arguments to the standard output. Display the ARGs on the standard output followed by a newline. Options: -n do not append a newline -e enable interpretation of the following backslash escapes -E explicitly suppress interpretation of backslash escapes echo' interprets the following backslash-escaped characters:
- \a
- alert (bell)
- \b
- backspace
- \c
- suppress further output
- \e
- escape character
- \f
- form feed
- \n
- new line
- \r
- carriage return
- \t
- horizontal tab
- \v
- vertical tab
- \
- backslash
- \0nnn
- the character whose ASCII code is NNN (octal). NNN can be 0 to 3 octal digits
- \xHH
- the eight-bit character whose value is HH (hexadecimal). HH can be one or two hex digits
- Exit Status:
- Returns success unless a write error occurs.
- Define local variables.
- Create a local variable called NAME, and give it VALUE. OPTION can be any option accepted by declare'. Local variables can only be used within a function; they are visible only to the function where they are defined and its children. Exit Status: Returns success unless an invalid option is supplied, an error occurs, or the shell is not executing a function. Set variable values and attributes. Obsolete. See **help declare'**.
- Set variable values and attributes.
- Declare variables and give them attributes. If no NAMES are given, display the attributes and values of all variables.
- Options:
- -f
- restrict action or display to function names and definitions
- -F
- restrict display to function names only (plus line number and source file when debugging)
- -g
- create global variables when used in a shell function; otherwise ignored
- -p
- display the attributes and value of each NAME
- Options which set attributes:
- -a
- to make NAMES indexed arrays (if supported)
- -A
- to make NAMES associative arrays (if supported)
- -i
- to make NAMES have the integer' attribute -l to convert NAMES to lower case on assignment -r to make NAMES readonly -t to make NAMES have the **trace'** attribute
- -u
- to convert NAMES to upper case on assignment
- -x
- to make NAMES export
- Using + ' instead of - ' turns off the given attribute.
- Variables with the integer attribute have arithmetic evaluation (see

- the `let` command) performed when the variable is assigned a value.
- When used in a function, `declare` makes NAMES local, as with the `local` command. The `-g` option suppresses this behavior.
- **Exit Status:**
- Returns success unless an invalid option is supplied or an error occurs.
- Execute a simple command or display information about commands.
- Runs COMMAND with ARGS suppressing shell function lookup, or display information about the specified COMMANDs. Can be used to invoke commands on disk when a function with the same name exists.
- **Options:**
- `-p`
- use a default value for PATH that is guaranteed to find all of the standard utilities
- `-v`
- print a description of COMMAND similar to the `type` builtin
- `-V` print a more verbose description of each COMMAND
- Exit Status:
 - Returns exit status of COMMAND, or failure if COMMAND is not found.
 - Return an unsuccessful result.
 - Exit Status:
 - Always fails.
 - Return a successful result.
 - Exit Status:
 - Always succeeds.
 - Null command.
 - No effect; the command does nothing.
 - Exit Status:
 - Always succeeds.
 - Print the name of the current working directory.
 - Options:
 - `-L` print the value of \$PWD if it names the current working directory
 - `-P` print the physical directory, without any symbolic links
 - By default, `pwd` behaves as if `-L` were specified.
 - Exit Status:
 - Returns 0 unless an invalid option is given or the current directory cannot be read.
 - Change the shell working directory.
 - Change the current directory to DIR. The default DIR is the value of the `HOME` shell variable.
 - The variable `CDPATH` defines the search path for the directory containing `DIR`. Alternative directory names in `CDPATH` are separated by a colon (:).
 - A null directory name is the same as the current directory. If `DIR` begins with a slash (/), then `CDPATH` is not used.
 - If the directory is not found, and the shell option `cdablevars` is set,
 - the word is assumed to be a variable name. If that variable has a value,
 - its value is used for `DIR`.
 - Options:
 - `-L`
 - force symbolic links to be followed
 - `-P`
 - use the physical directory structure without following symbolic
 - `-e`
 - if the `-P` option is supplied, and the current working directory cannot be determined successfully, exit with a non-zero status
 - The default is to follow symbolic links, as if `-L` were specified.
 - Exit Status:
 - Returns 0 if the directory is changed, and if \$PWD is set successfully when `-P` is used; non-zero otherwise.
 - Return the context of the current subroutine call.
 - Without `EXPR`, returns "`$line $filename`". With `EXPR`, returns "`$line $subroutine $filename`"; this extra information can be used to provide a stack trace.
 - The value of `EXPR` indicates how many call frames to go back before the current one; the top frame is frame 0.
 - Exit Status:
 - Returns 0 unless the shell is not executing a shell function or `EXPR` is invalid.
 - Execute shell builtins.
 - Execute `SHELL-BUILTIN` with arguments ARGS without performing command lookup. This is useful when you wish to reimplement a shell builtin as a shell function, but need to execute the builtin within the function.
 - Exit Status:
 - Returns the exit status of `SHELL-BUILTIN`, or false if `SHELL-BUILTIN` is not a shell builtin.
 - Resume for, while, or until loops.
 - Resumes the next iteration of the enclosing `FOR`, `WHILE` or `UNTIL` loop.
 - If `N` is specified, resumes the `N`th enclosing loop.
 - Exit Status:
 - The exit status is 0 unless `N` is not greater than or equal to 1.
 - Exit for, while, or until loops.
 - Exit a `FOR`, `WHILE` or `UNTIL` loop. If `N` is specified, break `N` enclosing loops.
 - Exit Status:
 - The exit status is 0 unless `N` is not greater than or equal to 1.
 - Set Readline key bindings and variables.
 - Bind a key sequence to a Readline function or a macro, or set `a`.
 - Readline variable. The non-option argument syntax is equivalent to that found in `~/.inputrc`, but must be passed as a single argument:
 - e.g., `bind "\C-x\C-r": re-read-init-file'.`
 - Options:
 - `-m` keymap Use `KEYMAP` as the keymap for the duration of this command. Acceptable keymap names are `emacs`, `emacs-standard`, `emacs-meta`, `emacs-ctlx`, `vi`, `vi-move`, `vi-command`, and `vi-insert`.
 - `-l` List names of functions.
 - `-P` List function names and bindings.
 - `-p` List functions and bindings in a form that can be reused as input.
 - `-S` List key sequences that invoke macros and their values
 - `-s` List key sequences that invoke macros and their values in a form that can be reused as input.
 - `-V` List variable names and values
 - `-v` List variable names and values in a form that can be reused as input.
 - `-q` function-name Query about which keys invoke the named function.
 - `-u` function-name Unbind all keys which are bound to the named function.
 - `-r` keyseq Remove the binding for `KEYSEQ`.
 - `-f` filename Read key bindings from `FILENAME`.
 - `-x` keyseq:shell-command Cause `SHELL-COMMAND` to be executed when `KEYSEQ` is entered.
 - Exit Status:
 - bind returns 0 unless an unrecognized option is given or an error occurs.
 - Remove each `NAME` from the list of defined aliases.
 - Options:
 - `-a` remove all alias definitions.
 - Return success unless a `NAME` is not an existing alias.
 - Define or display aliases.
 - Without arguments, `alias` prints the list of aliases in the reusable
 - **form** `alias NAME=VALUE` on standard output.
 - Otherwise, an alias is defined for each `NAME` whose `VALUE` is given.
 - A trailing space in `VALUE` causes the next word to be checked for alias substitution when the alias is expanded.
 - Options:
 - `-p` Print all defined aliases in a reusable format
 - Exit Status:
 - alias returns true unless a `NAME` is supplied for which no alias has been defined.
 - alias `[-p] [name[=value] ...]`
 - unalias `[-a] name [name ...]`
 - bind `[-lpvsPVS] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command]`
 - break `[n]`
 - continue
 - continue `[n]`
 - builtin `[shell-builtin [arg ...]]`
 - caller
 - caller `[expr]`
 - cd `[-l] [-P [-e]] [dir]`
 - pwd `[-LP]`
 - command `[-pvv] command [arg ...]`
 - declare
 - declare `[-aAfFgIltRux] [-p] [name[=value] ...]`
 - typeset
 - typeset `[-aAfFgIltRux] [-p] name[=value] ...`
 - local `[option] name[=value] ...`
 - echo `[-neE] [arg ...]`
 - enable
 - enable `[-a] [-dnps] [-f filename] [name ...]`
 - eval `[arg ...]`
 - getopt `optstring name [arg]`
 - exec `[-cl] [-a name] [command [arguments ...]] [redirection ...]`
 - exit `[n]`
 - logout `[n]`
 - fc `[-e ename] [-lnr] [first] [last] or fc -s [pat=rep] [command]`
 - fg `[job_spec]`
 - bg `[job_spec ...]`
 - hash `[-lr] [-p pathname] [-dt] [name ...]`
 - help `[-dms] [pattern ...]`
 - history `[-c] [-d offset] [n] or history -anrw [filename] or history -ps arg [arg...]`
 - jobs `[-lnprs] [jobspec ...] or jobs -x command [args]`
 - disown
 - disown `[-h] [-ar] [jobspec ...]`
 - kill `[-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]`
 - let `arg [arg ...]`
 - read `[-ers] [-a array] [-d delim] [-i text] [-n nchars] [-N nchars] [-p prompt] [-t timeout] [-u fd]`
 - return
 - return `[n]`
 - set `[-abefhknmptuvxBCHP] [-o option-name] [--] [arg ...]`
 - unset `[-f] [-v] [name ...]`
 - export
 - export `[-fn] [name[=value] ...] or`

```
export -p</li> <li>readonly</li> <li>readonly [-aAf] [name=value] ...] or readonly -p</li> <li>shift [n]</li> <li>source</li> <li>source filename
[arguments]</li> <li>. filename [arguments]</li> <li>suspend</li> <li>suspend [-f]</li> <li>test [expr]</li> <li>[ arg... ]</li> <li>trap [-lp] [[arg]
signal_spec ...]</li> <li>type [-afptP] name [name ...]</li> <li>ulimit</li> <li>ulimit [-SHacdefilmnpqrstuvx] [limit]</li> <li>umask [-p] [-S] [mode]
</li> <li>wait [id]</li> <li>for NAME [in WORDS ... ] ; do COMMANDS; done</li> <li>for (( exp1; exp2; exp3 )); do COMMANDS; done</li> <li>select NAME
[in WORDS ... ;] do COMMANDS; done</li> <li>time [-p] pipeline</li> <li>case WORD in [PATTERN [| PATTERN]...) COMMANDS ;;)... esac</li> <li>if
COMMANDS; then COMMANDS; [ elif COMMANDS; then COMMANDS; ]... [ else COMMANDS; ] fi</li> <li>while COMMANDS; do COMMANDS; done</li> <li>until
COMMANDS; do COMMANDS; done</li> <li>coproc [NAME] command [redirections]</li> <li>function name { COMMANDS ; } or name () { COMMANDS ; }</li> <li>{
... }</li> <li>job_spec [&]</li> <li>(( ... ))</li> <li>(( expression ))</li> <li>[[ ... ]]</li> <li>[[ expression ]]</li> <li>variables</li>
<li>variables - Names and meanings of some shell variables</li> <li>pushd [-n] [+N | -N | dir]</li> <li>popd [-n] [+N | -N]</li> <li>dirs [-clpv] [+N]
[-N]</li> <li>shopt [-pgsu] [-o] [optname ...]</li> <li>printf</li> <li>printf [-v var] format [arguments]</li> <li>complete</li> <li>complete [-
abdefghjksuv] [-pr] [-DE] [-o option] [-A action] [-G globpat] [-W wordlist] [-F functio</li> <li>compgen</li> <li>compgen [-abdefghjksuv] [-o option]
[-A action] [-G globpat] [-W wordlist] [-F function] [-C comma</li> <li>compopt</li> <li>compopt [-o|+o option] [-DE] [name ...]</li> <li>mapfile</li>
<li>mapfile [-n count] [-O origin] [-s count] [-t] [-u fd] [-C callback] [-c quantum] [array]</li> <li>readarray</li> <li>readarray [-n count] [-O
origin] [-s count] [-t] [-u fd] [-C callback] [-c quantum] [array]</li> <li>alias </li> <li>%s: invalid alias name
```

- *line editing not enabled*
- *bind*builtin
- *lpVPsSf:q:u:m:r:x:*
- %s': invalid keymap name %s: cannot read: %s %s': unknown function name
- %s is not bound to any keys.
- %s can be invoked via
- "%s"%s
- %s': cannot unbind only meaningful in a <code>for',</code>while', or until' loop
- loop count
- %s %s %s
- OLDPWD=
- HOME not set
- OLDPWD not set
- CDPATH
- /bin/usr/bin/sbin:/usr/sbin
- *command*builtin
- *line %d:*
- %s: *usage:*
- %s: *numeric argument required*
- %s: *not found*
- %s: *invalid option name*
- *invalid octal number*
- *invalid hex number*
- %s: *invalid signal specification*
- %s': not a pid or valid job spec %s: %s out of range argument %s out of range %s: no such job %s: no job
control no job control %s: restricted %s: not a shell builtin write error: %s error setting terminal
attributes: %s error getting terminal attributes: %s %s: error retrieving current directory: %s: %s %s: ambiguous job
spec SIGJUNK Unknown %2d) %s +acfgilprtuxAF cannot use -f to make functions
- %s %d %s
- %s: cannot destroy array variables in this way
- %s: cannot convert associative to indexed array
- can only be used in a function
- *adnpsf:*
- *enable %s*
- *enable -n %s*
- cannot open shared object %s: %s
- cannot find %s in shared object %s: %s
- %s: not dynamically loaded
- %s: cannot delete: %s
- %s: not a regular file
- %s: file is too large
- evalfile
- *parseandexecute top*
- *pedispose*
- *parseandexecute*
- *parsestring top*
- *parsestring*
- %s: cannot execute: %s
- ~/bashlogout
- /etc/bash/bashlogout
- *There are stopped jobs.*
- *There are running jobs.*
- *not login shell: use* *exit'* logout \${FCEDIT:-\${EDITOR:-ed}} \${FCEDIT:-\${EDITOR:-vi}} :e:lnrs no command
found history specification bash-fc %s: cannot open temp file: %s fc builtin current job %d
started without job control builtin hash -p %s %s hashing disabled dlp:rt command %s: hash table
empty %s: cannot open: %s These shell commands are defined internally. Type *help'* to see this list.
- *Type* *help name'* to find out more about the function *name'*.
- *Use* *info bash'* to find out more about the shell in general. Use <code>man -k' or</code>info' to find out more about commands not in this
list. A star (*) next to a name means that the command is disabled. Shell commands matching keyword

- **Shell commands matching keywords** ` %%s - SYNOPSIS DESCRIPTION SEE ALSO %%sbash(1) IMPLEMENTATION no help topics match <code>%%s'. Try</code>help help' or <code>man -k %%s' or</code>info %%s'. acd:npsrw cannot use more than one of -anrw %%s: history expansion failed history position %%5d%c %%s no other options allowed with -x'`
- **lpnxrs**
- **jobsbuiltin**
- **Unknown error**
- **(%ld) - %s**
- **%%s: arguments must be process or job IDs**
- **MAPFILE**
- **%%s: invalid file descriptor specification**
- **%d: invalid file descriptor: %s**
- **%%s: invalid line count**
- **%%s: invalid array origin**
- **%%s: invalid callback quantum**
- **u:n:O:tC:c:s:**
- **internal error: getting variable name**
- **empty array variable name**
- **%%s: not an indexed array**
- **%%s %d %s**
- **directory stack empty**
- **directory stack index**
-
- **%2d %s**
- **%2d %s**
- **no other directory**
- **dirstack**
- **%%s: invalid timeout specification**
- **ersa:d:i:n:p:t:u:N:**
- **readbuiltin**
- **read error: %d: %s**
- **can only return'** from a function or sourced script editing-mode set %co %s allexport SHELLOPTS cannot simultaneously unset a function and a variable %%s: cannot unset %%s: cannot unset: readonly %s %%s: not an array variable braceexpand errexit errtrace functrace hashall histexpand interactive-comments keyword monitor noclobber noexec noglob notify nounset onecmd physical pipefail privileged xtrace declare -%s %%s -%s %%s: not a function shift count filename argument required %%s: file not found cannot suspend cannot suspend a login shell unknown trap -- %s %d trap -- %s %s %%s is aliased to %s'
- **alias %s=%s**
- **%%s is a shell keyword**
- **%%s is a function**
- **%%s is a shell builtin**
- **%%s is %s**
- **%%s is hashed (%s)**
- **(%s, -%c)**
- **(-%c)**
- **%-20s %16s**
- **unlimited**
- **%%s: cannot get limit: %s**
- **%c': bad command %%s: cannot modify limit: %s core file size blocks data seg size kbytes scheduling priority file size pending signals max locked memory max memory size open files pipe size 512 bytes POSIX message queues real-time priority stack size cpu time seconds max user processes virtual memory file locks u=%s,g=%s,o=%s %%c: invalid symbolic mode operator**
- **%c': invalid symbolic mode character umask%%s OPTARG autocd %%s: invalid shell option name shopt %%s BASHOPTS cannot set and unset shell options simultaneously cdable_vars cdspell checkhash checkjobs checkwinsize cmdhist compat31 compat32 compat40 compat41 direxpend dirspell dotglob execfail expand_aliases extdebug extglob extquote failglob force_ignores globstar gnu_errfmt histappend histreedit histverify hostcomplete huponexit interactive_comments lastpipe lithist mailwarn noemptycmd_completion nocaseglob nocasematch nullglob progcomp promptvars restricted_shell shift_verbose sourcepath xpg_echo missing hex digit for \x missing unicode digit for \%c warning: %s: %s #'-+ 0 hjlLtZ %%s: missing format character**
- **%c': invalid time format specification %%c: invalid format character**
- **%%s: illegal option -- %c**
- **%%s: option requires an argument -- %c**
- **bashdefault**
- **abcdefghijklmnopqrstuvwxyzA:G:W:P:S:X:F:C:DE**
- **%%s: invalid action name**
- **complete**
- **-o %s**
- **default**
- **dimnames**
- **filenames**

- *nospace*
- *plusdirs*
- *-A %s*
- *arrayvar*
- *binding*
- *enabled*
- *helptopic*
- *hostname*
- *running*
- *setopt*
- *signal*
- *%s %s*
- *%s: no completion specification*
- *compopt*
- *+o %s*
- *warning: -F option may not work as you expect*
- *warning: -C option may not work as you expect*
- *not currently executing completion function*
- *directory*
- *service*
- *xdigit*
- *backspace*
- *vertical-tab*
- *form-feed*
- *carriage-return*
- *exclamation-mark*
- *quotation-mark*
- *number-sign*
- *dollar-sign*
- *percent-sign*
- *ampersand*
- *apostrophe*
- *left-parenthesis*
- *right-parenthesis*
- *asterisk*
- *plus-sign*
- *hyphen*
- *hyphen-minus*
- *period*
- *full-stop*
- *semicolon*
- *less-than-sign*
- *equals-sign*
- *greater-than-sign*
- *question-mark*
- *commercial-at*
- *left-square-bracket*
- *backslash*
- *reverse-solidus*
- *right-square-bracket*
- *circumflex*
- *circumflex-accent*
- *underscore*
- *grave-accent*
- *left-brace*
- *left-curly-bracket*
- *vertical-line*
- *right-brace*
- *right-curly-bracket*
- *BASHREMATCH*
- */dev/fd/*
- */dev/std*
- *%s: bad network path specification*
- *socket*
- *connect*
- *%ldm%d.%03ds*
- *shmakepath*
- *shrealpath*
- *TMPDIR*
- */var/tmp*
- */usr/tmp*
- *%s/%s-%lu*
- *%s/cur*

- %s/tmp
- %s/new
- invalid base
- 0123456789abcdef
- 0123456789ABCDEF
- .shstrtab
- .interp
- .note.ABTag
- .gnu.hash
- .dynsym
- .dynstr
- .gnu.version
- .gnu.versionr
- .rela.dyn
- .rela.lA64.pltoff
- .rodata
- .lA64.unwindinfo
- .lA64.unwind
- .initarray
- .fini_array
- .data.rel.ro
- .dynamic
- .ctors
- .dtors
- .sdata

UTF-8 Strings

No detected.

UTF-16LE Strings

- fh#kPp
- <=%?>
- +,-./<
- #\$\$%&'123+,-./:
- ;\$\$%&'
- 67@AEw
- !XYZ[
- #\$\$x%&'()*+,-./012

Extracted URLs

- http://gnu.org/licenses/gpl.html

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 4.94
- **Status:** This file likely contains Standard Text or human-readable text.

VirusTotal Analysis

Scan Date: 2024-12-19 01:22:45

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 62
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 1
- Type-unsupported: 13

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

| Engine Name | Detection Category | Detection |
|-------------|--------------------|------------|
| No results | No results | No results |
